

Exact and finite presieving in explicit Chen bounds: a GRH threshold and quantitative distribution interfaces

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Abstract

We study the gap between explicit Chen bounds and the verified binary Goldbach range. We combine exact weights for the smallest primes, finite Rosser weights for medium primes, and Rosser weights for larger primes; truncated Brun weights provide an alternative construction. Finite first-failure masses use exponents selected per degree and nonnegative tilted squares on a band supplied by the preceding prefix. Positive prefix blocks retain preceding checks. A six-state count bounds the union of accepted supports; an integer roughness bound and a finite analytic interval reduce the GRH threshold. The actual Rosser support is also smaller than its defining level. A product condition uniform at every upper endpoint is used throughout; the problematic fixed-endpoint-only application in earlier candidates is not needed. The composite weights retain coefficients of absolute value at most one and have support below Q_*H_0D , where H_0 bounds the medium-prime support and Q_* contains only the exactly presieved primes, avoiding a full primordial cost through T . Using published explicit sieve constants and the GRH AP and bilinear estimates of Bordignon–Starichkova, we obtain $N \geq \exp(24274)$ with normalized surplus exceeding 10^{-5} , and the stronger surplus 0.0181 for $N \geq \exp(31000)$. Quantitative signed distribution inputs at levels $3/4$, $9/10$, $19/20$, and $99/100$ give parameterized thresholds without RH or GRH, retaining unspecified constants and starting points. A quantified distribution input starting at $4 \cdot 10^{18}$ is withdrawn after an integrality obstruction. This unreviewed manuscript states its numerical certificates and analytic dependencies separately.

1 Statements, normalization, and comparison

Put $L = \log N$, and let Ω count prime factors with multiplicity. Write $\pi_2(N)$ for the number of primes $p < N$ for which $N - p \geq 2$ and $\Omega(N - p) \leq 2$. Define

$$C_\infty = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right), \quad C_N = C_\infty \prod_{\substack{p>2 \\ p|N}} \frac{p-1}{p-2}, \quad U_N = 2e^{-\gamma}C_N.$$

Since the product over *all* integers $n \geq 3$ of $1 - (n-1)^{-2}$ equals $1/2$, $C_N > 1/2$ and $U_N > e^{-\gamma}$. The finite certificate also gives $C_\infty > 0.66$, hence the stronger uniform lower bound $U_N > U_* := 1.32e^{-\gamma}$ used below. This normalization follows the actual sieve product; it is not the $2e^\gamma C_N$ printed in the accessible GRH preprint.

Theorem 1 (GRH threshold). *Assume GRH for Dirichlet L -functions, including RH for ζ . Every even integer $N \geq e^{24274}$ satisfies*

$$\pi_2(N) > 10^{-5} \frac{U_N N}{\log^2 N} > 0.$$

For $N \geq e^{31000}$ the coefficient 10^{-5} can be replaced by 0.0181.

The published GRH result [6] has threshold $e^{e^{14}}$; the older preprint [5] has $e^{e^{15.85}}$. Here $\log(24274) = 10.097161097\dots$. The new threshold has about 10543 decimal digits, compared with about 522284 for the published threshold. Theorem 1 is a result of this unreviewed manuscript, not a published record. The stronger earlier candidates e^{8500} and e^{15000} remain unaudited at their original sieve interface and are not asserted by this theorem. We do not identify the internal unconditional manuscript at $e^{e^{24.9}}$ with an independently audited published result.

Let $B = 4 \cdot 10^{18}$. The following *rejected candidate*, which was previously proposed as an additional hypothesis, is retained for audit:

$$\mathcal{E}_{9/10}(x) := \sum_{d \leq x^{9/10}} \mu^2(d) \max_{(a,d)=1} \left| \pi(x; d, a) - \frac{\pi(x)}{\varphi(d)} \right| \leq \frac{x}{10 \log^2 x} \quad (x \geq B). \quad (\text{QEH}_B)$$

Remark 2 (Withdrawn full-range claim). *The former full-range claim based on (QEH_B) is withdrawn. The hypothesis itself is false at B , as proved below. Positive sieve-budget arithmetic under an inconsistent premise is not a meaningful conditional bridge.*

Ordinary Elliott–Halberstam is a standard asymptotic conjecture, with constants depending on the distribution level and logarithmic power. It does not provide this particular finite starting point or constant. Conversely, a single-level, single-logarithmic-power finite condition would not imply all of ordinary EH. They must not be described as two logically ordered versions of the same conjecture. The original published Chen theorem assumes GRH alone, and the GRH theorem above keeps that same conjectural assumption. RH for the zeta function is included in GRH here. EH and GEH are separate recognized distribution conjectures: for example, the GPY, Maynard, and Polymath conditional prime-gap results in [12] explicitly invoke them. GRH is not known to imply either conjecture. A hypothesis with prescribed numerical constants and a prescribed finite starting point must be identified separately; the standard asymptotic notation does not supply those data. Thus a distribution estimate with unspecified constants can be a consequence of standard EH, while the same estimate with prescribed constants and a prescribed starting point is additional quantitative input. The four distribution routes below use EH-style input in place of GRH, rather than requiring GRH together with EH.

Proposition 3 (Integrality obstruction). *For $2 \leq H < x$, put $P = \pi(x)$. Then*

$$\sum_{d \leq H} \mu^2(d) \max_{(a,d)=1} \left| \pi(x; d, a) - \frac{P}{\varphi(d)} \right| \geq \frac{\pi(H)}{2} - \sqrt{x}.$$

Consequently (QEH_B) fails at $x = B$ unconditionally.

Proof. For $x < 3$ the lower bound is negative and the assertion is immediate. We may therefore assume $x \geq 3$, so $P - 1 \geq 1$. Restrict the sum to prime moduli $q \leq H$, and put $m = q - 1$. The integer counts over reduced classes sum to $P - 1$, because the prime q itself is omitted. Write $P = km + r$, $0 \leq r < m$. If $r = 0$, some class has count at most $k - 1$, so its error is at least one. If $2 \leq r < m$, there are classes with count at most k and at least $k + 1$, so the maximum error is at least $\max(r/m, 1 - r/m) \geq 1/2$. The remaining case $r = 1$ requires $q - 1 \mid P - 1$. There are at most $\tau(P - 1) \leq 2\sqrt{P - 1} < 2\sqrt{x}$ such exceptional prime moduli. This proves the lower bound. At B , take $H = B^{9/10}$ and $J = \lfloor H \rfloor$. By [7, Lemma 3.1], $\pi(J) > J/\log J$ for $J \geq 71$. The Arb certificate gives

$$\frac{J}{2 \log J} - \sqrt{B} > 7.15819 \cdot 10^{14}, \quad \frac{B}{10 \log^2 B} < 2.18026 \cdot 10^{14}.$$

The lower bound already exceeds the proposed upper bound by a factor greater than 3.28. The earlier candidate with level 3/4 and budget $B/\log^4 B$ also fails this check, by a factor greater than 1.169. Both are rejected, independently of RH or GRH. $\square$

Proposition 4 (Distribution constants and onsets). *Suppose $0 < \theta < 1$ and $\mathcal{E}_\theta(x) \leq Cx/\log^4 x$ for every $x \geq X$, where $C > 0$, $X > 1$, and the complete square-free modulus sum is defined in (34). For $y \geq \log X$ with $e^{\theta y} \geq 72$, necessarily*

$$C > \mathfrak{I}_\theta(y), \quad \mathfrak{I}_\theta(y) = \frac{y^3}{2\theta} (e^{-(1-\theta)y} - e^{-y}) - y^4 e^{-y/2}. \quad (1)$$

Let $R = \max\{\log X, A, \sqrt{bC}\}$, where $A > 0$ and $b > 0$. If $\mathfrak{I}_\theta(M) > M^2/b$ and $e^{\theta M} \geq 72$, then $R > M$. The following are strict, certified necessary bounds, with $B = 4 \cdot 10^{18}$:

θ	C if $X \leq B$	$\log X$ if $C = 1$	b	R
3/4	> 1.1697	> 43	20	> 22
9/10	> 602.35	> 143	10	> 57
19/20	> 4858.17	> 336	10	> 130
99/100	> 25860.94	> 2246	10	> 834

These are necessary constraints, not sufficient choices of C, X . The witnesses are not asserted to give optimal lower limits.

Proof. Put $x = e^y$, $H = x^\theta$, and $J = \lfloor H \rfloor \geq 71$. Proposition 3 and the prime-count bound in [7, Lemma 3.1] give

$$\mathcal{E}_\theta(x) > \frac{J}{2 \log J} - \sqrt{x} > \frac{H-1}{2 \log H} - \sqrt{x}.$$

Multiplying by y^4/e^y gives (1). For the endpoint constraint, suppose $R \leq M$. Then $e^M \geq X$, so $C > \mathfrak{I}_\theta(M) > M^2/b \geq R^2/b \geq C$, a contradiction. The first numerical column evaluates (1) at $y = \log B$. The second uses the witnesses $y = 43, 143, 336, 2246$, where $\mathfrak{I}_\theta(y) > 1$. The last uses $M = 22, 57, 130, 834$, respectively. The 192-bit Arb certificate checks each strict inequality, including its domain condition, and compares the closed formula with the separately evaluated normalized prime-count lower bound. $\square$

In particular, the level 99/100 thresholds in Theorem 23 are always greater than e^{834} under their present budget, even though their displayed analytic starts are 141 and 142. If $C = 1$, the same distribution input instead requires $X > e^{2246}$. The other fixed analytic starts still dominate the corresponding last-column bounds. A higher distribution level alone does not justify a lower effective threshold.

2 Product estimates and an unused fixed-endpoint candidate

For $t > 2$, use strict upper endpoints and put

$$D(t) = \log t \prod_{2 < p < t} \left(1 - \frac{1}{p-1}\right), \quad D_\infty = 2e^{-\gamma} C_\infty.$$

The identity

$$\left(1 - \frac{1}{p-1}\right) = \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{(p-1)^2}\right)$$

explains the normalization and avoids bounding two cancelling terms independently. We use the following two primary inputs:

$$\sum_{p < t} \frac{1}{p} \geq \log \log t + M - \frac{2.964 \cdot 10^{-6}}{\log t} \quad (t \geq 2), \quad (2)$$

$$\left| e^\gamma \log t \prod_{p < t} (1 - 1/p) - 1 \right| \leq \delta(t) := \frac{3 \log t}{8\pi\sqrt{t}} \quad (t \geq 23.8, \text{ RH}). \quad (3)$$

These are respectively [7, Lemma 3.2] and [8, Corollary 1.3]. The strict endpoint in the second follows by a one-sided limit of the published non-strict version.

Let $T = 10^6$ and set

$$r_T = \frac{2.964 \cdot 10^{-6}}{\log T} + \frac{1}{T-1} + \frac{0.54}{T} < 1.754543 \cdot 10^{-6}.$$

Lemma 5. *The certificate described below proves $D(u) < D_\infty$ for $11 < u \leq T$. For $u > T$, $D(u)/D_\infty < e^{r_T}$. Consequently, under RH, for every $z > T$ and $1 < u < z$,*

$$\prod_{\substack{u \leq p < z \\ p \geq 11}} \left(1 - \frac{1}{p-1}\right)^{-1} < \frac{e^{r_T} \log z}{1 - \delta(z) \log u}. \quad (4)$$

In particular the factor preceding $\log z / \log u$ is less than $1 + 2 \cdot 10^{-6}$ for $z \geq e^{40}$, and less than $1 + 1/700$ for $z \geq B^{1/3}$.

Proof. Between consecutive primes $D(u)$ is increasing, and its downward jumps occur just after a prime. It is enough to check $D(p)$ for prime $11 < p \leq T$ and the endpoint T . Let $C_T = \prod_{2 < p \leq T} (1 - (p-1)^{-2})$. Since

$$\log \frac{C_T}{C_\infty} \leq \sum_{p > T} \frac{1}{p(p-2)} < \frac{1}{T-1},$$

we may compare every peak with the rigorous lower bound $2e^{-\gamma} C_T e^{-1/(T-1)}$. The script `conditional_chen_product_certificate.py` uses a complete Eratosthenes sieve and 192-bit Arb arithmetic. It checks 78493 peaks; the largest occurs at $p = 463181$. Its ball lies below 0.741284513, whereas the lower bound for D_∞ exceeds 0.741307533. These are strict ball comparisons, not rounded floating-point decisions.

For $p > T$, put $r(p) = -\log(1 - 1/(p-1)) - 1/p$. The factorization above gives

$$0 < r(p) \leq \frac{1}{p(p-2)} + \frac{0.54}{p^2}.$$

Writing $E(u) = \sum_{p < u} 1/p - \log \log u - M$ gives the exact identity

$$\log \frac{D(u)}{D_\infty} = -E(u) + \sum_{p \geq u} r(p).$$

Use (2) and bound the tails by sums over all integers larger than T . This proves the claimed r_T bound. Also (3) and $C(z) \geq C_\infty$ give $D(z)/D_\infty \geq 1 - \delta(z)$. For $u > 11$, the product in (4) equals $D(u) \log z / (D(z) \log u)$. For $u \leq 11$, take a limit $u' \downarrow 11$ from above; no factor changes and $\log u' \geq \log u$. Finally $\delta(z)$ decreases for $z > e^2$; the two endpoint comparisons are certified in the scripts. $\square$

Deleting primes dividing N , or reducing any local density below $1/(p-1)$, only decreases the left side. We take $\mathcal{P}_N$ to be the odd primes not dividing N and preserve its primes at most 11. The actual Q_N divides 1155, hence $\log Q_N < 7.1$. We occasionally use the weaker bound $\log Q_N < 8$.

Remark 6 (Endpoint quantifiers). *Lemma 5 is a condition at the specified upper endpoint z , for every smaller u . It is not a uniform condition for all intermediate upper endpoints: at $u = 17, v = 19$ the normalized product equals $\frac{16 \log 17}{15 \log 19} > 1.0263$. The wording of [4, Theorem 6], also reproduced in [9, Lemma 2.2], is a fixed-upper-endpoint condition. However, the proof of BJS Lemma 11 applies an induction at lower upper endpoints, through the recursion $T_n(D, z) = \sum_{p < z} g(p) T_{n-1}(D/p, p)$ in its relevant ranges. The product hypothesis at such an endpoint p does not follow simply by dividing two inequalities at z . We have not verified an alternative argument that closes this gap for the present product. This is a pending audit of our application, not a claim that the published theorem is false. Neither a positive budget nor a finer parameter scan settles this interface. The earlier candidates using only this lemma remain pending. Theorem 1 instead uses the uniform product estimate and finite preserving proved next, so its induction has the required premise at every intermediate endpoint.*

3 Uniform products and composite sieve weights

For an integer $T \geq 10^4$, put

$$b_T = \max \left\{ \frac{2}{\sqrt{T} \log T}, \frac{6.836 \cdot 10^{-6}}{\log(10^{12})} \right\}, \quad K_T = \exp \left(b_T + \frac{2.964 \cdot 10^{-6}}{\log T} + \frac{1}{T-1} + \frac{0.54}{T} \right).$$

Lemma 7 (Uniform product condition). *For every $1 < u < v$, unconditionally,*

$$\prod_{\substack{u \leq p < v \\ p > T}} \left(1 - \frac{1}{p-1} \right)^{-1} \leq K_T \frac{\log v}{\log u}.$$

Deleting primes or decreasing local densities preserves this bound.

Proof. BJS Lemma 16 gives $E(v) \leq 2/(\sqrt{v} \log v)$ for $v \leq 10^{12}$; JS Lemma 3.2 gives $E(v) \leq 6.836 \cdot 10^{-6}/\log v$ for $v > 10^{12}$. Strict endpoints follow by limits, so $E(v) \leq b_T$ for $v > T$. For $T < u < v$, use

$$\log \frac{D(u)}{D(v)} = E(v) - E(u) + \sum_{p \geq u} r(p) - \sum_{p \geq v} r(p) \leq b_T + \frac{2.964 \cdot 10^{-6}}{\log T} + \frac{1}{T-1} + \frac{0.54}{T}.$$

This proves the bound. For $u \leq T < v$, take $u' \downarrow T$ from above and use $\log u' \geq \log u$. For $v \leq T$ the product is empty. The argument includes all intermediate upper endpoints. $\square$

Lemma 8 (Finite ordered products with an analytic tail). *Let $10^4 \leq T < S \leq 10^{12}$ be even integers, put $d(t) = \log D(t)$, and let λ_Q be the explicit lower bound*

$$\lambda_Q = \log 2 - \gamma + \sum_{2 < p \leq Q} \log(1 - (p-1)^{-2}) + \log(1 - 1/Q) \leq \log D_\infty \quad (Q \geq 3).$$

Suppose a finite certificate supplies A_T, M_T such that

$$A_T \geq \sup_{T \leq u < v \leq S} (d(u) - d(v)), \quad M_T \geq \sup_{T \leq u \leq S} (d(u) - \lambda_Q).$$

Writing b_S, K_S for the original bounds above, set

$$B_T = \max\{A_T, M_T + b_S, \log K_S\}.$$

Then Lemma 7 holds with K_T replaced by e^{B_T} . It remains unconditional and uniform at every intermediate endpoint.

Proof. The prime-only tail of C_∞ is at least the product over all integers $n > Q$:

$$\prod_{n>Q} (1 - (n-1)^{-2}) = (Q-1)/Q.$$

This proves the asserted lower bound λ_Q . For $T < u < v \leq S$, use the bound A_T . For $T < u \leq S < v$, the identity for $D(v)/D_\infty$ gives $d(v) - \log D_\infty = -E(v) + \sum_{p \geq v} r(p) \geq -b_S$, so $d(u) - d(v) \leq M_T + b_S$. For $S \leq u < v$, apply the original uniform bound with cutoff S ; S is composite, so it introduces no missing boundary prime. The cases $u \leq T$ and $v \leq T$ follow as in Lemma 7. Finally, $D(u)/D(v)$ is the product in that lemma multiplied by $\log u / \log v$.

The function $d(t)$ increases strictly between primes and has a negative jump at each prime. Thus its ordered drawdown on $[T, S]$ is bounded by checking every left limit as a potential earlier maximum and every right limit as a potential later minimum, together with the initial endpoint and the final endpoint. For adjacent finite intervals with peak, trough and drawdown enclosures (P_1, Q_1, A_1) and (P_2, Q_2, A_2) , the combined drawdown is bounded by $\max\{A_1, A_2, P_1 - Q_2\}$. This gives a finite, complete certificate for the continuum of ordered pairs, rather than a sample of individual endpoints. $\square$

In the applications below take $S = 10^8$, $Q = 10^7$, and use $\hat{K}_T = e^{B_T}$ from Lemma 8. The complete integer sieve generates all 5761454 odd primes through S and checks 5760226 prime jumps above the smallest cutoff. Strict Arb comparisons certify the following rational upper bounds:

T	$10^{12} \log \hat{K}_T$	$\hat{K}_T - 1$ (upper bound)
10000	987086120	0.000987573450
22000	582114805	0.000582284267
70000	291898888	0.000291941495
5000000	24260241	0.000024260536

Here the middle column is a certified upper bound, not an asserted exact maximum. Finite and infinite ranges are joined by the three cases in the proof; no conjectural prime-product estimate is used.

Let $g(p) = 1/(p-1)$ on the odd primes not dividing N . Choose $2 \leq w_0 < T < z$ and split $P_N(z)$ into the exact part P_E with $p \leq w_0$, the medium-prime part P_0 with $w_0 < p \leq T$, and the large-prime part P_1 with $T < p < z$. Put

$$Q_0 = P_E, \quad Q_* = \prod_{2 < p \leq w_0} p,$$

$$V_*^{(0)} = \prod_{2 < p \leq w_0} \left(1 - \frac{1}{p-1}\right), \quad W_*^{(0)} = \sum_{2 < p \leq w_0} \frac{1}{p-1}.$$

Then $Q_0 \mid Q_*$. Exact inclusion-exclusion on Q_0 has divisor sum $1_{(n, Q_0)=1}$ and support at most Q_* . Fix an even integer $k \geq 2$. Define Brun coefficients $b^+(d) = \mu(d)$ for $d \mid P_0$, $\omega(d) \leq k$, and zero otherwise; define $b^-(d)$ with cutoff $k-1$. The binomial identity $\sum_{j=0}^k (-1)^j \binom{r}{j} = \binom{r-1}{k}$ for $r \geq 1$

proves the pointwise upper bound, with the analogous negative identity for $k-1$ proving the lower bound. Put

$$V_0 = \prod_{p|P_0} (1 - g(p)), \quad B^\pm = \sum_{d|P_0} b^\pm(d)g(d), \quad \Delta = B^+ - B^-.$$

Then $B^- \leq V_0 \leq B^+$ and

$$0 \leq \Delta = \sum_{\substack{d|P_0 \\ \omega(d)=k}} g(d) \leq W_{w_0,T}^k/k!, \quad B^+ \leq (1 + \eta)V_0, \quad \Delta \leq \eta V_0,$$

provided

$$\begin{aligned} W_{w_0,T} &= \log \log T + 0.262 - \frac{1}{2} + c_0 + b_T - W_*^{(0)}, \\ c_0 &= \sum_{2 < p \leq 1000} \frac{1}{p(p-1)} + \frac{1}{1000}, \\ \eta &\geq \frac{V_*^{(0)} \log T e^{b_T} W_{w_0,T}^k}{U_* k!}. \end{aligned} \tag{5}$$

The inequalities $B^- \leq V_0 \leq B^+$ follow by averaging the pointwise Brun inequalities over independent prime-divisibility indicators with probabilities $g(p)$. Their difference is the nonnegative elementary symmetric sum of degree k displayed above. Indeed $M < 0.262$, and the integer telescoping tail bounds the correction from $1/p$ to $1/(p-1)$ by c_0 . The exact identity for $D(T+)$ gives the complete odd-prime product through T at least $U_* e^{-b_T} / \log T$. Dividing this product by $V_*^{(0)}$ gives a lower bound for the complete medium-prime product, hence

$$V_0 \geq \frac{U_* e^{-b_T}}{V_*^{(0)} \log T}.$$

The medium density sum is at most $W_{w_0,T}$. These two bounds remain valid when primes dividing N are omitted, including omitted primes in the exact part: the bounds use the complete medium-prime product and sum, not $V(P_E)$ as a substitute for $V_*^{(0)}$. The bound $M < 0.262$ is independently certified by the partial sum $\gamma + \sum_{p \leq 1000} (\log(1 - 1/p) + 1/p)$, whose omitted terms are negative. The bound $C_\infty > 0.66$ is certified by the finite product through 10^4 , times $e^{-1/9999}$.

There is a second medium-prime construction that uses the divisor product rather than its number of prime factors to restrict support. It needs no dimension-one product premise on the medium primes.

Lemma 9 (Finite Rosser presieving). *Let $D_0 \geq T^2$ and $\sigma > 0$. On P_0 , take $b^\pm(d) = \mu(d)$ if, for $d = p_1 \cdots p_n$ with $p_1 > \cdots > p_n$, every odd prefix $m \leq n$ (upper sign), or every even prefix (lower sign), satisfies $p_1 \cdots p_m p_m^2 < D_0$; otherwise take zero. These weights have support $d < D_0$, coefficients in $\{\mu(d), 0\}$, and pointwise divisor sums bracketing $1_{(n,P_0)=1}$. With $B^\pm, V_0, \Delta$ defined as above, put*

$$\mathcal{R}_{w_0,T}(D_0, \sigma) = D_0^{-\sigma} \sum_{\substack{w_0 < r \leq T \\ r \text{ prime}}} \frac{g(r)}{1 - g(r)} r^{3\sigma} \prod_{\substack{r < p \leq T \\ p \text{ prime}}} \frac{1 + g(p)p^\sigma}{1 - g(p)}, \quad g(p) = \frac{1}{p-1}. \tag{6}$$

The sum and product in this expression use all medium odd primes, including any primes dividing N . Then

$$0 \leq \Delta \leq \mathcal{R}_{w_0,T}(D_0, \sigma) V_0, \quad V_0 \leq B^+ \leq (1 + \mathcal{R}_{w_0,T}(D_0, \sigma)) V_0.$$

Proof. For any finite prime set, expand its full inclusion–exclusion product. Partition the terms omitted by b^+ according to their first failing odd prefix, and those omitted by b^- according to their first failing even prefix. Summing all possible remaining primes below the minimum prime r of that prefix gives $V_0(r) = \prod_{p|P_0, p < r} (1 - g(p))$. If $T_n^{(0)}$ is the sum of $g(p_1 \cdots p_n) V_0(p_n)$ over first failing prefixes of length n of the relevant parity, this proves the finite polynomial identities

$$B^+ = V_0 + \sum_{n \text{ odd}} T_n^{(0)}, \quad B^- = V_0 - \sum_{n \text{ even}} T_n^{(0)}, \quad \Delta = \sum_{n \geq 1} T_n^{(0)}.$$

They are the finite version of the identities used in BJS Proposition 13. They also hold when $g(p) = 1_{p|n}$: all terms are nonnegative, giving the pointwise brackets. To see the support statement, if the last prefix has the checked parity then $dp_n^2 < D_0$; otherwise, for $n \geq 2$, the preceding checked prefix gives $d < p_1 \cdots p_{n-1} p_{n-1}^2 < D_0$. The remaining case is the one-prime lower weight, where $p \leq T < D_0$.

Every first failure satisfies $dp_n^2 \geq D_0$. Drop the preceding prefix restrictions, apply $1_{dp_n^2 \geq D_0} \leq (dp_n^2/D_0)^\sigma$, and group by $r = p_n$. Since

$$\frac{g(d)V_0(r)}{V_0} = \frac{g(r)}{1 - g(r)} \prod_{p|d, p > r} g(p) \prod_{p|P_0, p > r} (1 - g(p))^{-1},$$

summing every subset of primes larger than r gives exactly the positive expression in (6) for the actual prime set. There is no factor two: the odd and even sums partition the prefix lengths in Δ . Restoring deleted primes adds nonnegative summands and factors $(1 + g(p)p^\sigma)/(1 - g(p)) \geq 1$, so the complete-prime expression dominates it. Finally $B^+ - V_0 \leq \Delta$. $\square$

Lemma 10 (Separate finite gaps). *Write $a_0 = (B^+ - V_0)/V_0$ and $b_0 = (V_0 - B^-)/V_0$. For each medium prime r , let*

$$H_r(t) = \prod_{r < p \leq T} (1 + g(p)p^\sigma t), \quad e_j(r) = [t^j]H_r(t), \quad W_r = \prod_{r < p \leq T} (1 - g(p))^{-1}.$$

Both products use all medium primes. At $D_0 = T^u$, a prefix with k primes larger than its minimum r satisfies $dp_{\min}^2 \leq T^k r^3$. Retain only degrees for which $T^k r^3 \geq T^u$. If $u > 3$, also retain only $k \geq \lfloor u - 2 \rfloor$. Define E_r and O_r as the sums of $e_k(r)$ over the retained even and odd degrees, respectively. Then

$$a_0 \leq R_+ := D_0^{-\sigma} \sum_{w_0 < r \leq T} \frac{g(r)}{1 - g(r)} r^{3\sigma} W_r E_r, \quad b_0 \leq R_- := D_0^{-\sigma} \sum_{w_0 < r \leq T} \frac{g(r)}{1 - g(r)} r^{3\sigma} W_r O_r.$$

The bounds remain valid upon deleting primes dividing N . All terms and elementary coefficients are nonnegative.

Proof. Apply the first-failure proof above separately to odd and even prefix lengths. An odd prefix has an even number of primes larger than r , and conversely for an even prefix. The degree restriction from $T^k r^3 < D_0$ removes only impossible failures. When $u > 3$, a prefix of length n with $n + 2 \leq u$ cannot fail: for $n = 1$ its product is at most $T^3 < T^u$, and for $n \geq 2$ distinct primes make the comparison strict even at equality $n + 2 = u$. Thus $k = n - 1 \geq \lfloor u - 2 \rfloor$. Each retained polynomial has positive coefficients; restoring primes increases it and W_r , and adds positive minimum-prime terms. For rational $u = A/B$, the degree test is exactly $r^{3B} T^{kB} \geq T^A$, so no approximate cutoff is required.

The finite certificate maintains low elementary coefficients and the two tails beyond a fixed degree $M \geq \lceil u \rceil$. On adding a larger prime with $h = g(p)p^\sigma$, the low coefficients update by $e'_j = e_j + he_{j-1}$ in descending order. If M is odd, the tails update by $E' = E + hO$ and $O' = O + h(E + e_{M-1})$; if M is even they update by $E' = E + h(O + e_{M-1})$ and $O' = O + hE$, using old values throughout. Adding the required low coefficients to these tails evaluates E_r, O_r using positive arithmetic only. $\square$

Lemma 11 (Local exponents for separate degrees). *Let $\Sigma \subset [0, \infty)$ be finite and nonempty, let $D_0 = T^u$ with $u > 3$, and let $M \geq \lceil u \rceil$ be an integer. For each medium prime r , partition the retained even and odd degrees of Lemma 10 into singleton degrees below M and their respective residual tails at or above M . For such a group J and $\sigma \in \Sigma$, put*

$$q_{r,\sigma}(J) = D_0^{-\sigma} \frac{g(r)}{1 - g(r)} r^{3\sigma} W_r \sum_{k \in J} [x^k] \prod_{r < p \leq T} (1 + g(p)p^\sigma x).$$

Then

$$a_0 \leq \widehat{R}_+ := \sum_r \sum_{J \text{ even}} \min_{\sigma \in \Sigma} q_{r,\sigma}(J), \quad b_0 \leq \widehat{R}_- := \sum_r \sum_{J \text{ odd}} \min_{\sigma \in \Sigma} q_{r,\sigma}(J).$$

The expressions use complete prime sets and remain valid when primes dividing N are deleted. Their evaluation uses the positive recurrence of Lemma 10 separately for each exponent.

Proof. Fix r and a degree group J . Its contribution to the normalized first-failure sum is independent of σ . For every $\sigma \geq 0$, the inequality $1_{dr^2 \geq D_0} \leq (dr^2/D_0)^\sigma$ bounds that contribution by $q_{r,\sigma}(J)$; the exponent zero is also permitted. Every member of the finite family is thus an upper bound for the same nonnegative contribution, so their minimum is an upper bound. Sum over the disjoint groups and minimum primes. Since $M \geq \lceil u \rceil$, all degrees in the residual tails satisfy the necessary degree conditions; no possible failure is discarded. For a deleted prime set the complete-prime expression dominates its corresponding contribution for each exponent. Taking a minimum preserves this domination. Neither the sieve weights nor their support have changed; only their finite mass estimates have been tightened. $\square$

Lemma 12 (Nonnegative quadratic product bounds). *Fix a minimum prime r and one of the degree groups J in Lemma 11. For $\sigma > 0$, use the complete larger prime set and put*

$$m_j = \sum_{\substack{S \subset \{p:r < p \leq T\} \\ |S| \in J}} \left(\prod_{p \in S} g(p) \right) \left(\frac{r^3 \prod_{p \in S} p}{D_0} \right)^{j\sigma}, \quad j = 0, 1, 2.$$

For every $0 \leq b < 1$, the normalized first-failure contribution of this group is at most

$$\frac{g(r)}{1 - g(r)} W_r \frac{m_2 - 2bm_1 + b^2m_0}{(1 - b)^2}.$$

If $m_2 < m_1 < m_0$, $m_0 - 2m_1 + m_2 > 0$, and $b_* = (m_1 - m_2)/(m_0 - m_1) \in (0, 1)$, one may use the bound

$$\frac{g(r)}{1 - g(r)} W_r \frac{m_0 m_2 - m_1^2}{m_0 - 2m_1 + m_2}.$$

Take a finite minimum with the ordinary Rankin bounds and sum over the same degree groups and minimum primes. Denote the resulting gap bounds by $\widehat{R}_+^{(2)}$ and $\widehat{R}_-^{(2)}$. They remain valid when primes dividing N are deleted.

Proof. For $X \geq 0$ and $0 \leq b < 1$, $1_{X \geq 1} \leq (X - b)^2 / (1 - b)^2$: the right side is nonnegative everywhere and at least one on $X \geq 1$. Use $X = (r^3 \prod_{p \in S} p / D_0)^\sigma$. Dropping preceding-prefix restrictions adds only nonnegative polynomial values. For a deleted prime set, restoring primes increases both the subset sum of these nonnegative values and W_r , so the displayed complete-prime bound dominates the actual contribution for every fixed b . Only after this domination do we optimize b ; no monotonicity of a moment quotient under deletion is assumed. Differentiating the rational quadratic gives b_* and the second displayed expression. Finite weighted Cauchy-Schwarz gives $m_0 m_2 - m_1^2 \geq 0$. The moments at exponents $0, \sigma, 2\sigma$ are evaluated by the positive coefficient and parity-tail recurrences above. Interval subtraction and division enclose the quadratic expression; the optimizer is used only when all required signs are certified strictly. An unresolved comparison is skipped. Enclosing the maximum of the variance expression and zero is legitimate by its proved nonnegativity, even when a rounded enclosure of exact zero has a slightly negative lower endpoint. The weights and support are unchanged. $\square$

Lemma 13 (A preceding-pass product band). *Let $D_0 = T^u$ with $u > 3$. Every first-failing prefix d , with minimum prime r , satisfies*

$$D_0 / r^2 \leq d < D_0.$$

Consequently $\delta = \log(dr^2 / D_0)$ belongs to $[0, R]$, where $R = 2 \log r$. For $\sigma, c, b \geq 0$, the function

$$P_{\sigma, c, b}(\delta) = \exp(\sigma \delta) \{1 + c\delta + b\delta(R - \delta)\}^2$$

is nonnegative on the real line and at least one on $[0, R]$. For a retained degree group J , take the complete larger prime set and put

$$m_j = \sum_{\substack{S \subset \{p: r < p \leq T\} \\ |S| \in J}} \left(\prod_{p \in S} g(p) \right) e^{\sigma \delta_S} \delta_S^j, \quad \delta_S = \log(r^3 \prod_{p \in S} p / D_0), \quad 0 \leq j \leq 4.$$

Write

$$v_0 = m_1, \quad v_1 = Rm_1 - m_2, \quad A = m_2, \quad B = Rm_2 - m_3, \quad C = R^2 m_2 - 2Rm_3 + m_4.$$

The group's normalized first-failure contribution is bounded by $g(r)W_r / (1 - g(r))$ times any of the following certified candidates: m_0 ; $m_0 - v_0^2 / A$ if $v_0 < 0$, $A > 0$; $m_0 - v_1^2 / C$ if $v_1 < 0$, $C > 0$; and $m_0 + v_0 c_ + v_1 b_*$ if*

$$AC - B^2 > 0, \quad c_* = \frac{Bv_1 - Cv_0}{AC - B^2} \geq 0, \quad b_* = \frac{Bv_0 - Av_1}{AC - B^2} \geq 0.$$

Take finite minima with ordinary Rankin candidates, then sum over the degree groups and r . Denote the resulting gap bounds by $\widehat{R}_\pm^{\text{band}}$. They remain valid upon deleting primes dividing N ; neither the weights nor their support changes.

Proof. A one-prime prefix cannot fail since $T^3 < D_0$. A two-prime prefix has product less than $T^2 < D_0$. For a longer first failure of length n , the preceding checked prefix of length $n - 2$ passed. Writing its last prime as s and the last two primes as $q > r$, we have $s > q > r$ and $d = d_{n-2}qr < d_{n-2}s^2 < D_0$. Failure gives $dr^2 \geq D_0$, proving the band. On the band both δ and $\delta(R - \delta)$ are nonnegative; this proves the majorant. Restoring primes increases the expectation of this nonnegative function and W_r , before any optimization. This proves domination for every fixed σ, c, b , including for deleted prime sets.

The complete expectation of the square equals $m_0 + 2cv_0 + 2bv_1 + c^2A + 2cbB + b^2C$. The two boundary candidates and the interior candidate follow by minimizing this quadratic over $c, b \geq 0$ when their stated signs hold. The matrix with entries A, B, C is a finite weighted Gram matrix, so all accepted minimum values are nonnegative. Interval maximum with zero therefore preserves enclosure. All optimizer conditions are checked strictly; unresolved candidates are omitted.

Raw logarithmic moments are computed positively. Put

$$E_{k,j} = \sum_{|S|=k} \prod_{p \in S} g(p) p^\sigma \frac{(\sum_{p \in S} \log p)^j}{j!}.$$

Adding a prime of logarithm ℓ updates $E_{k,j}$ by adding $g(p)p^\sigma \sum_{a=0}^j \ell^{j-a} E_{k-1,a} / (j-a)!$. The two parity tails use the same positive convolution on their opposite tail and the boundary degree, as in Lemma 10. After these positive recurrences, finite binomial shifts give the m_j with Arb inclusion. The signed shifts and Gram formulas are not evaluated by floating subtraction or unproved truncation. Native polynomial multiplication truncated after order four evaluates these exact moment coefficients: higher orders cannot enter them. Any fixed logarithm base greater than one gives the same majorant family after rescaling σ, c, b by positive constants. $\square$

Lemma 14 (Positive prefix blocks). *Use the finite Rosser weights of Lemma 9, with $D_0 = T^u$ and $u > 3$. Let $r_1 < r_2 < \dots$ be the complete medium primes, and let K be the largest integer with $r_1 \dots r_K < D_0$. Choose $0 < h < \log r_1$, put $J = \lceil \log D_0 / h \rceil$, and label a chosen prefix by $B = \sum_{p|d} \lfloor \log p / h \rfloor$. Partition the complete medium primes into descending blocks with common label $b = \lfloor \log p / h \rfloor$. In a block set*

$$c = \left\lceil \frac{\log D_0 - 3 \log p_{\min}}{h} \right\rceil, \quad f = \max \left\{ 0, \left\lceil \frac{\log D_0 - 3 \log p_{\max}}{h} \right\rceil - K + 1 \right\}.$$

For each of the upper and lower weights, keep two nonnegative polynomials X_0, X_1 for the parity of possible passing prefixes, truncated to degrees $B < J$, initially $(1, 0)$. Let A append label b , flip parity, and, when the appended prefix has the checked parity, retain only source labels $B < c$. Always truncate resulting labels to $B < J$. Let $\mathcal{F}$ sum source coefficients with $B \geq f$ in the parity that is checked upon appending one prime. For $t_p = 1/(p-2)$ form the polynomials in the operator variable a

$$D(a) = \prod_{p \text{ in block}} (1 + t_p + t_p a), \quad C(a) = \frac{D(a) - 1}{1 + a}.$$

Their coefficients are nonnegative. For each block add $\mathcal{F} \sum_{j=0}^K [a^j] C(a) A^j X$ to the corresponding gap bound, and replace X by $\sum_{j=0}^K [a^j] D(a) A^j X$. The resulting two totals bound $(B^+ - V_0)/V_0$ and $(V_0 - B^-)/V_0$ for every deletion of medium primes. The rounding changes only the mass bound, not the defining sieve weights.

Proof. Every accepted prefix has product below D_0 , by the support proof in Lemma 9; every first-failing prefix does also, by Lemma 13. Both therefore have at most K chosen primes. For a source prefix of degree k and true logarithm v , floor labels give $hB \leq v \leq h(B+k)$. A checked pass requires $v + 3 \log p < \log D_0$, so $B < c$. A first failure has $k \leq K-1$ and $v + 3 \log p \geq \log D_0$, so $B \geq f$. Thus these are necessary possible-pass and possible-failure conditions. A rounding ambiguity may contribute to both; this only increases the nonnegative bounds. Truncating labels at J removes no actual accepted prefix.

Before a prime p , multiply the accepted-prefix generating polynomials by the factor $W = \prod_{q>p}(1 - g(q))^{-1}$ over previously processed primes. For a common relaxed append operator A , processing p gives

$$X' = ((1 + t_p)I + t_p A)X, \quad G' = G + t_p \mathcal{F}X,$$

where G is the accumulated normalized gap. The actual passing prefixes and first failures are bounded by this positive evolution. The augmented update matrices commute because A and $\mathcal{F}$ are common throughout the block. Multiplying them gives $X' = D(A)X$ and $G' = G + \mathcal{F}C(A)X$. Equivalently, as a prime with value t is added,

$$D_{\text{new}} = (1 + t + ta)D, \quad C_{\text{new}} = (1 + t + ta)C + t.$$

Starting from $D = 1, C = 0$, these recurrences use positive arithmetic. With e_m the elementary symmetric sums of the block's t_p ,

$$[a^j]D = \sum_{m \geq j} \binom{m}{j} e_m, \quad [a^j]C = \sum_{m \geq j+1} \binom{m-1}{j} e_m.$$

An A^j term selects j primes within this block. Actual passing prefixes have total degree at most K , and actual first failures have at most $K - 1$ preceding selected primes. Hence omitting terms $j > K$ cannot remove an actual contribution. Artificial paths of greater total degree may remain; their positive mass only enlarges the bound.

For a deleted prime set use the same complete-set K , blocks, and cutoffs. They still contain every actual possible pass or failure. Deleting primes decreases every coefficient of D and C , as their displayed positive formulas show. The positive operators therefore give complete-set domination. This proves the uniform claim; no cancellation of prefix masses is used. $\square$

Lemma 15 (Sharper finite support). *For the complete medium prime set, let p_0 be its least member and put*

$$\kappa = \max \left\{ \frac{1}{p_0^2}, \max_{\substack{w_0 < r < s \leq T \\ r, s \text{ prime}}} \frac{r}{s^2} \right\},$$

with the inner maximum omitted for a singleton set. If $D_0 = T^u$, $u > 3$, every nonzero coefficient of b^+ or b^- has support $d < \kappa D_0$. This remains true for every deleted prime set. Consequently the composite weights below have support less than $\kappa Q_ D_0 D$. Their finite gap estimates still use the original level D_0 .*

Proof. If the final prefix is checked, $dr^2 < D_0$, hence $d < D_0/r^2 \leq \kappa D_0$. If it is unchecked and has length at least two, the preceding checked prefix has last prime $s > r$ and passed; thus $d = d_{\text{prev}} r < D_0 r/s^2 \leq \kappa D_0$. The remaining one-prime lower coefficient satisfies $d \leq T < \kappa T^u$, since $\kappa \geq 1/T^2$ and $u > 3$. The empty coefficient also satisfies the bound. The complete-set maximum dominates every pair in a deleted set. For a fixed s its largest r is the preceding complete prime, so a complete adjacent-prime scan computes the maximum. The convolution support follows by multiplying the exact, medium, and large support bounds. No coefficient or gap mass has changed. $\square$

For the present cutoffs the exact maxima are $11/169$ at $w_0 = 5$ and $5/49$ at $w_0 = 3$. The support-only certificate verifies these maxima and 52488 finite coefficient/deletion cases. The current endpoint charges below use $H_0 = \kappa D_0$. Their finite gap estimates still use the original defining level D_0 .

Take the usual Rosser coefficients $\lambda_1^\pm$ on P_1 at level D . They equal either $\mu(d)$ or zero, have support $d < D$, and their divisor sums bracket $1_{(n, P_1)=1}$; see the construction used in BJS Proposition 13 and [13, Theorem 9.3]. In particular, the upper divisor sum is nonnegative. A precise definition takes $d = p_1 \cdots p_r$, $p_1 > \cdots > p_r$, and requires $p_1 \cdots p_m p_m^2 < D$ at every odd $m \leq r$ for the upper weights and every even $m \leq r$ for the lower weights. Their main masses $G_1^\pm$ obey the BJS bounds with $Q = 1$, since either Lemma 7 or Lemma 8 verifies the product premise at every endpoint. Write $\mathcal{K}_T$ for the uniform product constant supplied by the chosen lemma.

Lemma 16 (Composite presieving). *Suppose $0 < \epsilon \leq 1/74$, $\mathcal{K}_T < 1 + \epsilon$, and use the BJS constants C_1, C_2 . Use either the Brun medium weights, with $H_0 = T^k$ and η as in (5), or the finite Rosser medium weights, with $H_0 = D_0$ (or $H_0 = \kappa D_0$ under Lemma 15) and $\eta \geq \mathcal{R}_{w_0, T}(D_0, \sigma)$. For a nonnegative finite sequence with local density g , write $s = \log D / \log z$ and $R = \sum_{d|P_N(z), d < Q_* H_0 D} |r(d)|$. If $D \geq z^2$ and $f(s) - \epsilon C_2 e^2 h(s) > 0$, then the lower bound in (7) holds. If $D \geq z$, its upper bound holds.*

Proof. First convolve the medium/large-prime coefficients to form

$$\tilde{\lambda}^+ = b^+ * \lambda_1^+, \quad \tilde{\lambda}^- = b^+ * \lambda_1^- - (b^+ - b^-) * \lambda_1^+.$$

For pointwise upper and lower brackets $a^- \leq I_0 \leq a^+$ and $c^- \leq I_1 \leq c^+$ with $a^+, c^+ \geq 0$, one has

$$a^- c^+ + a^+ c^- - a^+ c^+ \leq I_0 I_1 \leq a^+ c^+.$$

For the lower inequality, its left side is at most $I_0 c^+ + a^+ I_1 - a^+ c^+ = I_0 I_1 - (a^+ - I_0)(c^+ - I_1)$. Thus these convolutions are legitimate bounds for the medium/large parts. Multiply the pointwise inequalities by $1_{(n, Q_0)=1} \geq 0$, equivalently convolve both coefficients with $\mu_E(d) = \mu(d) 1_{d|Q_0}$, to obtain the full weights $\lambda^\pm$. Every divisor has a unique medium/large factorization. Write $b^\pm(d_0) = \mu(d_0) I^\pm$ and $\lambda_1^\pm(d_1) = \mu(d_1) J^\pm$, where all four indicators belong to $\{0, 1\}$. The normalized lower coefficient is $I^+ J^- - (I^+ - I^-) J^+$. If $J^+ = 0$, it is $I^+ J^-$; if $J^+ = 1$, it is I^- when $J^- = 1$ and $I^- - I^+$ otherwise. Thus it belongs to $\{-1, 0, 1\}$, as does the normalized upper coefficient. This argument applies to both medium constructions. The exact factorization is also unique and $|\mu_E| \leq 1$, so the full coefficients have the same bound. Their support is below $Q_* H_0 D$. Consequently no new divisor multiplicity is charged to R . Writing $V_E = \prod_{p|P_E} (1 - g(p))$, the upper main mass is $V_E B^+ G_1^+$ and is at most $(1 + \eta) V(z) (F(s) + \epsilon C_1 e^2 h(s))$. The lower main mass is $V_E (B^+ G_1^- - \Delta G_1^+)$ and is at least $V(z) (f(s) - \epsilon C_2 e^2 h(s) - \eta (F(s) + \epsilon C_1 e^2 h(s)))$. Here the positivity assumption permits using $B^+ \geq V_0$ in the first term. Expanding the divisor sums against the sequence and using the coefficient bounds proves the remainder claim. $\square$

The resulting inequalities are

$$\begin{aligned} S(A, z) &\geq |A| V(z) \{f(s) - \epsilon C_2 e^2 h(s) - \eta (F(s) + \epsilon C_1 e^2 h(s))\} - R, & D \geq z^2, \\ S(A, z) &\leq |A| V(z) (1 + \eta) \{F(s) + \epsilon C_1 e^2 h(s)\} + R, & D \geq z, \end{aligned} \quad (7)$$

where the lower bound includes the stated positivity condition, and $V(z) = \prod_{p|P(z)} (1 - g(p))$.

When $a_0 \leq \eta_+$ and $b_0 \leq \eta_-$ are supplied by Lemmas 10, 11, 12, 13, or 14, put $l(s) = f(s) - \epsilon C_2 e^2 h(s)$ and $u(s) = F(s) + \epsilon C_1 e^2 h(s)$. Since $u(s) \geq l(s)$ and $u(s) > 0$, the same composite weights give the sharper bounds

$$\begin{aligned} S(A, z) &\geq |A| V(z) \{(1 + \eta_+) l(s) - (\eta_+ + \eta_-) u(s)\} - R, & D \geq z^2, \\ S(A, z) &\leq |A| V(z) (1 + \eta_+) u(s) + R, & D \geq z. \end{aligned} \quad (8)$$

Indeed their normalized lower mass is at least $(1 + a_0)l - (a_0 + b_0)u = l - a_0(u - l) - b_0u$, which decreases in both a_0, b_0 ; the upper mass uses only a_0 . The earlier coarse form remains available without separate gap inputs. For the finite Rosser construction these refined inequalities use the separate inputs of Lemmas 10, 11, 12, 13, or 14 directly; the coarse constant can always be chosen as $\mathcal{R}_{w_0, T}(D_0, \sigma)$ and is not charged in this refined form.

4 Analytic inputs and elementary envelopes

In the ranges needed here,

$$F(s) = \frac{2e^\gamma}{s} \quad (1 \leq s \leq 3), \quad f(s) = \frac{2e^\gamma \log(s-1)}{s} \quad (2 \leq s \leq 4),$$

$$h(s) = e^{-2} \quad (1 \leq s \leq 2), \quad e^{-s} \quad (2 \leq s \leq 3), \quad 3e^{-s}/s \quad (s \geq 3).$$

For $3 \leq s \leq 4$, we also retain the exact expression from [4] (the explicit formulas for F, f), equivalently obtained from $(sF(s))' = f(s-1)$:

$$F(s) = \frac{2e^\gamma}{s} \left(1 + \int_3^s \frac{\log(x-2)}{x-1} dx \right). \quad (9)$$

The integral is evaluated by Arb/Acb analytic integration. Table 1 of [4] permits $(C_1, C_2) = (106, 108)$ when $\epsilon \leq 10^{-4}$, $(109, 110)$ when $\epsilon \leq 1/2000$, $(113, 114)$ when $\epsilon \leq 1/1000$, and $(116, 117)$ when $\epsilon \leq 1/700$.

The prime sequence is $A = \{N - p : p < N, p \text{ prime}, p \nmid N\}$, and $g(p) = 1/(p-1)$ on $\mathcal{P}_N$. The exact product identity and (3) imply

$$\frac{U_N}{\log x} \left(1 - \delta(x) - \frac{2L}{x} \right) < V(x) < \frac{U_N}{\log x} (1 + v), \quad (10)$$

for the lower bound whenever $x \geq B^{1/3}$, and for the upper bound at $x = N^{1/8}, N^{1/3}$ when $L \geq 8500$, with $v = 10^{-10}$. Indeed, the missing factors for $p \mid N, p \geq x$, have loss at most $\omega(N)/(x-1) \leq L/(\log 2(x-1)) < 2L/x$. The tail $C(x)/C_\infty$ is at most $\exp(1/(x-2))$. At both large endpoints its effect, the missing-factor loss, and $\delta(x)$ together are bounded by $Le^{-L/16} < v$.

Schoenfeld's RH bound, for the large x used here, is $|\pi(x) - \text{li}(x)| \leq \sqrt{x} \log x / (8\pi)$. For $t = \log x \geq 20$ elementary envelopes give

$$\frac{x}{t} (1 + 1/t) < \text{li}(x) < \frac{x}{t} (1 + 1/t + 3/t^2).$$

For completeness, the two inequalities are checked at $t = 20$ by Arb evaluation of $\text{Ei}(20)$; their derivatives differ from $1/\log x$ by $-2/t^3$ and $(t-9)/t^4$, respectively. Thus, after subtracting the primes dividing N ,

$$|A| > N/L \quad (N \geq B), \quad |A| \leq (1 + 2/L)N/L \quad (L \geq 8500). \quad (11)$$

The required error comparisons at B follow already from $L_B = 42.832826035\dots$ and $\omega(N) \leq L/\log 2$. Likewise, for every $x \geq y = N^{1/3}$ and $L \geq 8500$,

$$\pi(x) \leq (1 + 4/L)x/\log x. \quad (12)$$

The error term in the normalized prime count is at most $L^2 e^{-L/6} / (72\pi)$ and is absorbed by $1/L - 27/L^2$.

Under GRH, the AP estimate recorded in [5, Section 2], from Ernvall-Hytönen–Palojärvi, gives

$$|\pi(N; d, a) - \text{li}(N)/\varphi(d)| \leq q_G(N)\sqrt{N} \log N \quad (3 \leq d \leq \sqrt{N}, (a, d) = 1).$$

For $L \geq 8500$, $q_G(N) < 0.17$. To verify this uniformly, omit the only negative term of the formula in BS and put

$$q_+(w) = 0.165 + 12.683/w + 254.980/w^2 + 2607.854/w^3 + 11605.056/w^4 \\ + e^{-w/4}\{1.314w + 0.092 \log w + 60.883 + 8.250 \log w/w + 939.260/w\}.$$

Every positive term decreases for $w \geq 8500$; the certificate gives $q_+(8500) < 0.166496$. Set $\beta = 7/2$ and $H = \sqrt{N}/L^\beta$. Then

$$\sum_{\substack{d \leq H \\ (d, N)=1}} \mu^2(d)|r(d)| \leq 0.12 N/L^{5/2} =: R_A. \quad (13)$$

Indeed, $r(1) = 0$, and $d = 2$ is excluded because N is even. For $d \geq 3$, $\varphi(d) \geq 2$, so centering at $\pi(N)/\varphi(d)$ and deleting primes costs at most

$$\left(0.17 + \frac{1}{16\pi} + \frac{2e^{-L/2}}{\log 2}\right) \sqrt{N}L.$$

The coefficient is less than 0.195. Use the unconditional bound $\sum_{d \leq H} \mu^2(d) \leq 0.608H$ for $H \geq 10^9$, recorded in BS; the certificate verifies 0.608 times this coefficient is less than 0.12. Separately, counting at most $y = N^{1/3}$ prime moduli gives

$$\sum_{\substack{z \leq q < y \\ q \nmid N}} |r(q)| \leq 0.2 N^{5/6}L =: R_{\text{prime}}. \quad (14)$$

This avoids charging $\log H$ against the entire sum R_A . We also use the unconditional bound

$$\sum_{d \leq H} \frac{\mu^2(d)}{\varphi(d)} \leq 1.1 \log H \quad (H \geq 10^9) \quad (15)$$

recorded in [5, Section 2].

We will repeatedly use the following consequence of RH.

Lemma 17 (Weighted reciprocal sums). *For $L \geq 8500$, $z = N^{1/8}$, and all $x \geq z$, $|\sum_{p < x} 1/p - \log \log x - M| \leq \xi = L^{-2}$. For a nonnegative monotone function w on $[a, b] \subseteq [z, \infty)$,*

$$\sum_{a \leq p < b} \frac{w(p)}{p} \leq \int_a^b \frac{w(t)}{t \log t} dt + 2\xi \max(w(a), w(b)).$$

Proof. Use [8, Corollary 1.3] and $\delta(z) = 3Le^{-L/16}/(64\pi) < L^{-2}$. Stieltjes integration by parts bounds the difference by $\xi(w(a) + w(b) + \text{Var}(w))$. Both endpoint conventions satisfy the same bound by limits. $\square$

5 The three terms in Chen's inequality

Set

$$\begin{aligned} \alpha &= 4215/10^6, \quad t = 1/2 - \alpha, \quad D = N^t, \quad \epsilon = 1/41200, \quad z = N^{1/8}, \quad y = N^{1/3}, \\ s &= 4 - 8\alpha = 3.966280, \quad k = 8(1/6 - \alpha), \quad \delta = 10^{-8}, \quad \ell = \log(1 + \delta). \end{aligned}$$

The exact cutoff is $w_0 = 5$, so $Q_* = 15$ and $V_*^{(0)} = 3/8$. Use the finite Rosser medium weights with $T = 5 \cdot 10^6$, $D_0 = T^{49/8}$, $\eta_+ = 1/2768$, and $\eta_- = 1/2784$. Use Lemmas 11 and 13 with $M = 12$ and

$$\Sigma_G = \{0, 1/10, 7/50, 9/50, 11/50, 7/25, 9/25\}, \quad \Gamma_G = \{2/25, 3/25, 4/25, 1/5, 6/25\}.$$

The ordered uniform product certificate gives $\hat{K}_T - 1 < 2.426054 \cdot 10^{-5} < \epsilon$. Positive moment recurrences and certified shifted Gram combinations over all 348510 medium primes give $\hat{R}_+^{(\text{band})} < 0.000361187945 < \eta_+$ and $\hat{R}_-^{(\text{band})} < 0.000359148152 < \eta_-$, with the degree restrictions of Lemma 10. Lemma 15 gives $\kappa = 11/169$. For $L \geq 31000$, $\alpha L - (7/2) \log L > \log(\kappa Q_*) + (49/8) \log T$, increasing with L . The right side is less than 94.454; the certified support slack at the endpoint exceeds 0.015045. Therefore $\kappa Q_* D_0 D \leq H$. Also $D \geq z^2$, $D/y \geq z$, $D \geq y$, $y \leq H$, and $H \geq 10^9$. Every sieve in this section uses (8); its product premise holds uniformly at all intermediate endpoints.

Nathanson's combinatorial Chen inequality, reproduced in the preliminaries to sieving in [5], is

$$\pi_2(N) \geq S(A, z) - \frac{1}{2} \sum_{z \leq q < y} S(A_q, z) - \frac{1}{2} S(\mathcal{B}, y) - 2N^{7/8} - N^{1/3} - 1. \quad (16)$$

Here $\mathcal{B} = \{N - p_1 p_2 p_3 : z \leq p_1 < y \leq p_2 \leq p_3, p_1 p_2 p_3 < N, (p_1 p_2 p_3, N) = 1\}$. The extra subtraction of one safely excludes $N - p = 1$ under either convention for almost primes.

5.1 Lower sieve

From (8), (11), (10), and (13), in units $U_N N / L^2$,

$$\begin{aligned} S(A, z) / U_N N / L^2 &\geq M_0 - \frac{0.12}{U_* \sqrt{L}}, \\ M_0 &= 8(1 - v) \{ (1 + \eta_+) (f(s) - 108\epsilon e^2 h(s)) \\ &\quad - (\eta_+ + \eta_-) (F(s) + 106\epsilon e^2 h(s)) \}. \end{aligned} \quad (17)$$

5.2 The prime-factor sum

Apply the upper sieve to A_q with $D_q = D/q$. Primes $q \mid N$ contribute nothing and are omitted in all remainder sums below; main-term upper bounds may include them. Then $s_q = 8(t - \log q/L) \in [k, 3]$, so $F(s_q) = 2e^\gamma / s_q$. We retain the correction $e^2 h(s_q)$ rather than replacing it by one. Put $F_* = 2e^\gamma / k + 106\epsilon$ and $J = \log(8/3)$. Since both $F(s_q)$ and $h(s_q)$ increase with q , their sum is a nonnegative monotone weight, bounded by F_* . Lemma 17 and $1/(q-1) \leq (1 - z^{-1})^{-1}/q$ give

$$\sum_{z \leq q < y} \frac{F(s_q) + 106\epsilon e^2 h(s_q)}{q-1} \leq \frac{I + 106\epsilon J_h(t) + 2F_*/L^2}{1 - z^{-1}},$$

where direct integration yields

$$I = \int_{1/8}^{1/3} \frac{F(8(t - \beta))}{\beta} d\beta = \frac{e^\gamma}{4t} \log \left(6 \frac{3 - 8\alpha}{3 - 18\alpha} \right).$$

Writing $\beta_0 = t - 1/4$, which lies between $1/8$ and $1/3$, the two pieces of h give

$$J_h(t) = \int_{1/8}^{\beta_0} \frac{e^{2-8(t-\beta)}}{\beta} d\beta + \log \frac{1/3}{\beta_0} = \int_{1/8}^{1/3} \frac{e^{2h(8(t-\beta))}}{\beta} d\beta. \quad (18)$$

Thus $0 < J_h(t) < J$. At the fixed parameter value, analytic interval integration gives $J_h(t) = 0.715653586850489\dots$. In the error sum, $r_q(d) = r(qd) - r(q)/\varphi(d)$. The map $(q, d) \mapsto qd$, for $d \mid P(z)$ and $q \geq z$, is injective. Thus the total sieve remainder is at most $R_A + 0.55LR_{\text{prime}}$ by (15). The substitution $|A_q| = |A|/(q-1) + r(q)$ costs at most $(1+\eta_+)V(z)F_*R_{\text{prime}}$. It follows that

$$\frac{1}{2U_N N/L^2} \sum S(A_q, z) \leq M_1 + \frac{0.06}{U_* \sqrt{L}} + \frac{0.055L^4 e^{-L/6}}{U_*} + 0.8(1+\eta_+)(1+v)F_*L^2 e^{-L/6}, \quad (19)$$

$$M_1 = \frac{4(1+\eta_+)(1+2/L)(1+v)}{1-e^{-L/8}} \left(I + 106\epsilon J_h(t) + \frac{2F_*}{L^2} \right).$$

5.3 The switched triple-prime mass

Partition p_1 into the intervals $[w_j, (1+\delta)w_j)$, truncated at y , where $w_j = z(1+\delta)^j$. Enlarge the product restriction to $w_j p_2 p_3 < N$, and drop $(p_1, N) = 1$. Denote the disjoint enlarged sets by $\mathcal{B}_j$, with union $\bar{\mathcal{B}}$. Then $\mathcal{B} \subseteq \bar{\mathcal{B}}$ and its products are less than $(1+\delta)N$. Unique sorted prime factorization ensures that these are legitimate sets, including $p_2 = p_3$. Some enlarged differences $N - p_1 p_2 p_3$ can be negative. Translate each finite sequence by a sufficiently large integer multiple of $P_N(y)$ to make its indices positive. This preserves every divisibility condition $d \mid P_N(y)$, its cardinality, and all sieve remainders, so the arithmetic-function form of the cited sieve is still usable.

For $u = p_1$, put $w = \sqrt{(1+\delta)N/u}$ and

$$H(u) = \int_y^{\sqrt{N/u}} \frac{dv}{v \log v \log(N/uv)} = \frac{\log(2-3\beta)}{L(1-\beta)}, \quad \beta = \log u/L.$$

By (12), the mass is bounded by $(1+\delta)(1+4/L)N$ times $\sum_{z \leq p_1 < y} p_1^{-1} \sum_{y \leq p_2 < w} (p_2 \log(N/p_1 p_2))^{-1}$. The inner weight is increasing and at most $4/L$. Lemma 17 costs $8/L^3$; extending its integral from $\sqrt{N/u}$ to w costs at most $6\ell/L^2$. The function $H(u)$ is positive decreasing, with $H(z) < 1/L$, so the outer weighted sum costs at most $2/L^3$. Consequently

$$|\bar{\mathcal{B}}| \leq \frac{(1+\delta)(1+4/L)N}{L} \left\{ \bar{c} + \frac{2}{L^2} + \left(J + \frac{2}{L^2} \right) \left(\frac{6\ell}{L} + \frac{8}{L^2} \right) \right\}, \quad (20)$$

$$\bar{c} = \int_{1/8}^{1/3} \frac{\log(2-3\beta)}{\beta(1-\beta)} d\beta = 0.3630837292483046\dots < 0.363084.$$

The integral is certified by Arb/Acb analytic integration, with the logarithm's analyticity flag enabled. This direct argument uses a prime-count *upper* envelope and does not use the negative RH error printed in the triple-mass estimate of the old GRH preprint.

Apply the upper sieve at y to every $\mathcal{B}_j$ with level D . Its parameter $3t$ lies in $[1, 2]$. Therefore, using (20), the normalized half-main-term is at most

$$M_2 = \frac{(1+\eta_+)(1+v)}{2} \left(\frac{2e^\gamma}{t} + 318\epsilon \right) (1+\delta)(1+4/L) \left\{ \bar{c} + \frac{2}{L^2} + \left(J + \frac{2}{L^2} \right) \left(\frac{6\ell}{L} + \frac{8}{L^2} \right) \right\}. \quad (21)$$

6 Uniform rectangle errors and the half-line certificate

The GRH bilinear lemma in [5] states, for $|a(n)| \leq 1$, $X, Y, Z \geq 25$, $Z < Y$, $H \geq 10^9$, and

$$\left(\frac{\sqrt{Y} \log H}{q_G(Y) \log Y \log 2} \right)^{2/3} \leq Z,$$

that the total square-free AP error of $\sum_{n < X} \sum_{Z \leq p < Y} a(n)$ is at most

$$m(X, Y, H) \frac{XY \log^{5/3} H \log^{1/3} Y}{Y^{1/6}}. \quad (22)$$

The principal term here restricts $(np, d) = 1$. The explicit coefficient is

$$\begin{aligned} m = & 2.81q_G(Y)^{1/3} + \frac{2.809}{q_G(Y)^{2/3}\sqrt{Y}\log Y} + \frac{1.721\log^{2/3} H}{q_G(Y)^{1/3}Y^{2/3}\log^{2/3} Y \log \log H} \\ & + 5.498(X^{-1/2} + Y^{-1/2})Y^{1/6}\frac{\log^{1/3} H}{\log^{1/3} Y} + \frac{19.044H\log^{1/3} H}{XY^{5/6}\log^{1/3} Y}. \end{aligned}$$

For our rectangles, $X = N/w_j$, $Y = \min(y, (1 + \delta)w_j)$, $Z = w_j$, and $a(n)$ indicates $n = p_2 p_3$ with $y \leq p_2 \leq p_3$ and $(n, N) = 1$. Thus $|a(n)| \leq 1$, $X \geq N^{2/3}$, $Y \geq z$, $Y < \sqrt{N}$, and $XY \leq (1 + \delta)N$. The AP bound gives $0.16 < q_G(Y) < 0.64$ in this range; this follows also directly from the formula in [5, Section 2], since $\log Y \geq 1062.5$, its constant term is 0.165, and its only negative term is exponentially small. Termwise, without any monotonicity assertion for m , we have

$$\begin{aligned} m \leq & 2.81(0.64)^{1/3} + \frac{2.809e^{-L/16}}{(0.16)^{2/3}(L/8)} + 13e^{-L/12} \\ & + 9(e^{-5L/18} + e^{-L/24}) + 31e^{-13L/48} < 3. \end{aligned}$$

The rectangle condition follows from $13(1 + \delta)^{1/3}e^{-L/12} < 1$. Since both logarithms in (22) are at most $L/2$, the error in one rectangle is at most $\frac{3}{4}(1 + \delta)N^{47/48}L^2$.

Removing $(np_1, d) = 1$ from the principal term costs at most $2(1 + \delta)N\omega(d)/(z\varphi(d))$: a prime dividing d can divide either n or p_1 , and both possibilities are counted. Using $\omega(d) \leq \log H / \log 2$ and (15), its sum is at most $(1 + \delta)N^{7/8}L^2$, since $2 \cdot 1.1/(4 \log 2) < 1$ and $\log H \leq L/2$. There are $[5L/(24\ell)] < L/\ell$ rectangles. Thus the normalized half-remainder is bounded by

$$E_{\text{rect}}(L) = \frac{(1 + \delta)L^5}{2U_*\ell} \left(\frac{3}{4}e^{-L/48} + e^{-L/8} \right). \quad (23)$$

All endpoints, including the truncated final rectangle, have now been included; no floating-point underflow is used to discard errors.

Combining (16)–(23), set

$$\begin{aligned} E_{\text{AP}}(L) &= \frac{0.18}{U_*\sqrt{L}} + \frac{0.055L^4e^{-L/6}}{U_*} + 0.8(1 + \eta_+)(1 + v)F_*L^2e^{-L/6}, \\ E_{\text{fin}}(L) &= \frac{L^2}{U_*} \left(2e^{-L/8} + e^{-2L/3} + e^{-L} \right). \end{aligned}$$

Then

$$\pi_2(N)/U_N N/L^2 \geq M_0 - M_1(L) - M_2(L) - E_{\text{AP}}(L) - E_{\text{rect}}(L) - E_{\text{fin}}(L).$$

Every subtracted term is decreasing for $L \geq 31000$. For the exponential terms use $L^b e^{-aL}$ decreasing when $L > b/a$; the largest such cutoff is 240. All other factors are positive decreasing rational functions of L or $(1 - e^{-L/8})^{-1}$. The fixed α makes M_0 constant and avoids any moving parameter issue. At $L = 31000$, the 192-bit certificate gives

Term	certified value (rounded for display)
M_0	7.806859399895
M_1	6.498312561553
M_2	1.306394071261
E_{AP}	0.001379428851
E_{rect}	$< 10^{-248}$
E_{fin}	$< 10^{-1670}$
Surplus	> 0.0007733382290

This proves the earlier counting coefficient 0.0006, using the uniform product premise and composite weights established above. For its lower starting point, keep all finite weights and gap inputs fixed, but take $\alpha = 4238/10^6$ and $L_0 = 30824$. The same 192-bit checks of every support, parameter and remainder condition give support slack > 0.002084 and

$$M_0 - M_1 - M_2 - E_{\text{AP}} - E_{\text{rect}} - E_{\text{fin}} > 0.0000337669646 > 10^{-5}.$$

Here $E_{\text{rect}} < 10^{-248}$ and $E_{\text{fin}} < 10^{-1660}$; both are charged positively. The new $s = 3.966096$ and $k > 1$ obey all the same analytic ranges. With this fixed α the support slack increases and every deducted expense decreases on $L \geq 30824$, by the same monotonicity argument. Thus the earlier bound holds for $N \geq e^{30824}$, with $\log(30824) = 10.336048886\dots$ and about 13387 decimal digits at its starting point. Both calculations cover the entire half-line, rather than a finite mesh.

6.1 An enlarged finite grid and a sharper AP budget

The defining weights, T , D_0 , Q_* , M , Σ_G , and κ remain as above. Enlarge only the finite majorant choices to

$$\Gamma_G^{\text{new}} = \{a/25 : 2 \leq a \leq 9\}.$$

Lemma 13 applies to each of these nonnegative majorants; taking their finite minimum cannot increase either gap. A fresh complete integer sieve and 192-bit positive-moment recurrence over all 348510 medium primes gives

$$\widehat{R}_+ < 0.000349096808 < 1/2864, \quad \widehat{R}_- < 0.000347781454 < 1/2875.$$

Set $\epsilon = 1/41219$, which still exceeds the previously certified uniform product error, and use these two new gap budgets. With $\beta = 7/2$ and the previous AP charge, the endpoint already falls to $L_0 = 30779$.

There is a further saving in the AP charge. For a fixed $7/2 \leq \beta < 4$, set $H = \sqrt{N}/L^\beta$ and retain the positive envelope $q_+(L)$ above, rather than replacing it by 0.17. The proof of (13) gives

$$R_A \leq 0.608 P(L) \frac{N}{L^{\beta-1}}, \quad P(L) = q_+(L) + \frac{1}{16\pi} + \frac{2e^{-L/2}}{\log 2}.$$

Its coefficients in the lower sieve and half of the prime-factor sum total $3/2$. Hence replace only the leading term $0.18/(U_*\sqrt{L})$ in E_{AP} by

$$\frac{0.912 P(L)}{U_* L^{\beta-3}}. \tag{24}$$

Keep the two other AP charges, the entire rectangle charge, and every finite loss. The support condition becomes $\alpha L - \beta \log L > \log(\kappa Q_* D_0)$. All rectangle estimates above used only $H \leq \sqrt{N}$ and $H \geq 10^9$, and continue to hold.

That preceding threshold had $\log(30747) = 10.333547707\dots$ and about 13354 decimal digits. For the preceding assertions $N \geq e^{30747}$ with coefficient 10^{-5} and $N \geq e^{31000}$ with coefficient 0.001, use the following fixed parameters and strict certified endpoint inequalities:

	Lower starting point	Stronger count
L_0	30747	31000
α	$4255/10^6$	$4218/10^6$
β	$88/25$	$351/100$
s	3.965960	3.966256
Support slack	> 0.0005409294	> 0.0046277428
E_{AP}	< 0.0010576384	< 0.0011678640
E_{rect}	$< 10^{-248}$	$< 10^{-250}$
E_{fin}	$< 10^{-1659}$	$< 10^{-1673}$
Surplus	> 0.0000139212861	> 0.0010866414727

The certificate checks every preceding analytic range, the actual $H \geq 10^9$, and the support inequality at each new endpoint. For the half-line, $\alpha - \beta/L > 0$ and $1/2 - \beta/L > 0$ at the endpoint and thereafter. Thus support slack and H increase. The function $P(L)$ decreases for $L \geq 8500$, and $\beta > 3$; (24) therefore decreases. The remaining monotonicity arguments are unchanged. This proves those earlier assertions for all N in their respective half-lines. No distribution hypothesis beyond GRH has been introduced. The eight tested values $\beta = 3.50, 3.51, \dots, 3.57$ are a finite parameter search, not a proof of optimality.

6.2 All preceding checks and the actual modulus count

The following calculation proves the earlier starting point e^{26820} and the preceding stronger coefficient 0.0179. Use Lemma 14 with $T = 5000000$, $w_0 = 5$, $Q_* = 15$, $u = 87/16$, and $h = 1/200$. The complete integer sieve has 348510 medium primes; the exact degree bound is $K = 22$, the label limit is $J = 16775$, and there are 1810 nonempty blocks. The strict positive recurrences give

$$\hat{R}_+ < 0.000369501096 < 1/2706, \quad \hat{R}_- < 0.000369035200 < 1/2709.$$

Set $\epsilon = 1/41219$, $\eta_+ = 1/2706$, $\eta_- = 1/2709$, $(C_1, C_2) = (106, 108)$, and retain $\kappa = 11/169$. The actual support cost is

$$c = \log(\kappa Q_*) + u \log T = 83.849204066\dots$$

All uses of (8) accept the new separate gap inputs; its proof depends only on their domination of the two finite masses.

Lemma 18 (Counting the actual modulus support). *For the composite coefficients in Lemma 16, put $H_0 = \kappa D_0$ and let m be the number of complete exact primes. For every $\sigma > 0$, the number of their nonzero coefficients is at most*

$$2^m D H_0^\sigma \prod_{w_0 < p \leq T} (1 + p^{-\sigma}).$$

This bound holds for every deletion of primes dividing N and both signs. If $Q_*H_0D \leq H$, this number is at most νH , where

$$\nu = \frac{2^m}{Q_*} H_0^{\sigma-1} \prod_{w_0 < p \leq T} (1 + p^{-\sigma}). \quad (25)$$

Proof. Every contributing divisor has a unique exact, medium and large factorization. There are at most 2^m exact divisors. Every medium factor is square-free, uses only primes in $(w_0, T]$, and is below H_0 by Lemma 15, for either medium sign. Rankin's elementary inequality $1_{d_0 < H_0} \leq (H_0/d_0)^\sigma$ bounds their number by the Euler product in the statement. A large factor is a positive integer below D , so there are fewer than D choices. Counting all such triples bounds the coefficient support, even when some coefficients vanish. Deleting primes decreases the positive Euler product and the available exact factors, while the complete-set H_0 still bounds their support. Dividing by $Q_*H_0D \leq H$ proves the second assertion. $\square$

For $\sigma = 4/5$, a fresh 192-bit Euler-product sum over the complete medium primes gives

$$\log \prod_{5 < p \leq T} (1 + p^{-\sigma}) < 10.1391321865, \quad \nu < 0.000604653842 < 1/1653.$$

For the prime-factor sieves A_q , $q \geq z > T$ and the large level is D/q . For their aggregate AP remainder, write $d = ed_0d_1$ and absorb q into the large factor: $qd_1 < D$. Since $d \mid P(z)$, the prime q is the unique prime factor of qd that is at least z . Thus $(q, d) \mapsto qd$ is injective over the whole sum. All its images belong to the same enlarged set of exact, medium and large triples, so the same count bounds the aggregate first AP remainder. The additional centering term $r(q)/\varphi(d)$ retains its previous charge.

Set $H = \sqrt{N}/L^\beta$ with $\beta \geq 3$. The pointwise AP bound and centering argument used for (13) give, for each actual support,

$$R_{\text{support}} \leq \nu P(L) \frac{N}{L^{\beta-1}}, \quad P(L) = q_+(L) + \frac{1}{16\pi} + \frac{2e^{-L/2}}{\log 2}.$$

Thus the lower sieve and half of the prime-factor sum charge a total of $3R_{\text{support}}/2$. Replace (24) by

$$\frac{3\nu P(L)}{2U_*L^{\beta-3}}, \quad (26)$$

using the fixed upper budget $\nu_* = 1/1653$ in this expense. Both other AP terms, the prime-modulus centering cost, the full rectangle charge, and every finite loss remain in the proof. The bilinear lemma still uses the full maximum modulus H , rather than this count.

The strict endpoint certificate gives

	Lower starting point	Stronger count
L_0	26820	31000
α	4267/1000000	1853/500000
β	3	3
s	3.965864	3.970352
Support slack	> 0.0010264627	> 0.0115684828
E_{AP}	< 0.0002269656	< 0.0002268874
Surplus	> 0.000034241255	> 0.017948372354
E_{rect}	$< 10^{-212}$	$< 10^{-250}$
E_{fin}	$< 10^{-1446}$	$< 10^{-1673}$

The respective surpluses exceed 10^{-5} and 0.0179. All analytic ranges used in the earlier calculation are checked again, including $H \geq 10^9$, $T < z$, $3 < s < 4$, $k > 1$, and actual support $\alpha L - \beta \log L > c$. For each fixed parameter pair, support slack and H increase thereafter. With $\beta = 3$, the leading AP expense decreases with $P(L)$; every other expense decreases as before. The inequalities therefore hold on the whole half-lines. This uses GRH alone.

The previous prefix-block budget using the full square-free modulus count gave $N \geq e^{28279}$ and coefficient 0.0116 at e^{31000} . The original finite grid gave e^{28334} . The new support count accounts for the further improvement. The searched finite parameters and the four Rankin exponents do not prove global optimality.

6.3 Logarithmic support counts and rough large factors

Two elementary count bounds improve the previous support estimate. They introduce no distribution hypothesis. Throughout this subsection, the actual sieve coefficients remain those of Lemma 16.

Lemma 19 (Positive logarithmic-bin support count). *Let $\mathcal{M}$ be the complete medium-prime set, $H_0 = \kappa T^u > 1$, and let K be the largest number of its smallest primes whose product is below H_0 . Choose $0 < h < \log \min \mathcal{M}$, put $b(p) = \lfloor \log p/h \rfloor$ and $J = \lceil \log H_0/h \rceil$. Partition $\mathcal{M}$ into blocks $\mathcal{M}_b$ of common label b . For $\sigma \geq 0$, define positive coefficients by*

$$\prod_{p \in \mathcal{M}_b} (1 + p^{-\sigma} a) = \sum_j e_{b,j} a^j, \quad P(v) = \prod_b \left(\sum_{0 \leq j \leq K} e_{b,j} v^{jb} \right) \pmod{v^J}.$$

Write $P_B = [v^B]P$. The number of square-free products of primes in $\mathcal{M}$ that are below H_0 is at most

$$A_\sigma = \sum_{B < J} P_B \exp(\sigma \min\{\log H_0, h(B + K)\}). \quad (27)$$

The same complete-set upper bound holds after arbitrary prime deletion. For $\sigma = 0$, $e_{b,j} = \binom{|\mathcal{M}_b|}{j}$ and $A_0 = \sum_{B < J} P_B$.

Proof. A true product $d < H_0$ contains at most K primes. Its label B satisfies $hB \leq \log d < \log H_0$, hence $B < J$. Its block degrees are at most K , so its positive contribution $d^{-\sigma}$ occurs in P_B . Every intermediate label is at most its final label; truncation modulo v^J after each block cannot remove that contribution. Also $\log d \leq h(B + K)$, so its contribution to (27) is at least one. Products of excessive total degree that survive the block truncations are harmless additional positive contributions. Deleting primes decreases every elementary coefficient $e_{b,j}$, while the complete-set degree and label bounds remain valid. $\square$

Lemma 20 (Counting rough integers). *Let $V_T = \prod_{p \leq T} (1 - 1/p)$, and let η_+ be the complete normalized medium upper-gap budget supplied by Lemma 14 for $g(p) = 1/(p - 1)$. With the same D_0 , H_0 and exact odd product Q_* , the number $R_T(D)$ of positive integers $n < D$ having no prime factor at most T satisfies*

$$R_T(D) \leq a_T D + b_T, \quad a_T = (1 + \eta_+) V_T, \quad b_T = 2Q_* H_0. \quad (28)$$

Proof. Use full inclusion–exclusion at all primes at most w_0 , including 2, and the finite Rosser upper coefficients at $w_0 < p \leq T$. For the integer sequence the density is $g_0(p) = 1/p$. The normalized positive recurrences in Lemma 14 also hold with $t_p = g(p)/(1 - g(p))$: their passing and failing

operators depend only on the prime labels and D_0 , and every coefficient of $D(a)$ and $C(a)$ is a nonnegative polynomial in the t_p . Thus decreasing t_p decreases every propagated state and gap bound. Here $t_{0,p} = 1/(p-1) \leq 1/(p-2)$, so the normalized medium upper gap at density g_0 is at most η_+ . Multiplying by the exact small-prime factor gives an upper main mass $\sum_d \lambda_d^+ / d \leq a_T$.

These integer upper weights have $|\lambda_d^+| \leq 1$ and support $d < 2Q_*H_0$ by Lemma 15; the factor 2 is required for the exact even prime. The number of positive multiples of d strictly below D is $\lceil D/d \rceil - 1$, differing from D/d by an amount in $[-1, 0)$. Summing the pointwise upper weights therefore gives

$$R_T(D) \leq \sum_d \lambda_d^+ (\lceil D/d \rceil - 1) \leq D \sum_d \frac{\lambda_d^+}{d} + \#\{d : \lambda_d^+ \neq 0\} \leq a_T D + b_T.$$

This is an elementary integer-count bound, not an AP distribution input. $\square$

Put $m = \omega(Q_*)$ for the complete exact odd-prime set and

$$\nu_\sigma = \frac{2^m A_\sigma}{Q_* H_0}.$$

Every large-prime factor of a composite coefficient has no prime factor at most T . The coefficient support therefore has size at most $2^m A_\sigma (a_T D + b_T)$. If $Q_* H_0 D \leq H$, its size is at most

$$\nu_\sigma (a_T + b_T/D) H. \quad (29)$$

This counts all possible exact, medium and large factors; any vanishing coefficient only reduces the support. Prime deletion preserves the bound. For the aggregate A_q remainder, write $d = ed_0d_1$ with large level D/q and absorb q into d_1 . The resulting large factor $qd_1 < D$ has no prime factor at most T . Because $d \mid P(z)$ and $q \geq z$, q is the unique prime factor of qd that is at least z . The map $(q, d) \mapsto qd$ is therefore injective over the whole sum, and its complete image has at most $2^m A_\sigma R_T(D)$ elements. Thus (29) bounds the aggregate first AP remainder, without summing a separate floor charge for each q . The separate prime-modulus centering cost is retained.

For $T = 5000000$, $w_0 = 5$, $u = 87/16$ and $\kappa = 11/169$, use the preceding gap budgets and $\sigma = 0$, $h = 1/100$ for the count. The complete count uses 348510 medium primes, $K = 22$, $J = 8115$, and 992 nonempty blocks. Its positive polynomial calculation yields

$$A_0 < 5.488142 \cdot 10^{30}, \quad \nu_0 < 0.000008437984 < \nu_* := 1/118511.$$

The gap computation still uses $h = 1/200$ as in Section 6.2; the two bin widths serve different bounds. An independently regenerated complete integer prime set contains 348513 primes through T and gives

$$V_T < 0.036397999733, \quad a_T < a_* := 36411450583/10^{12}.$$

Finite analytic interval and the existing tail. Let $L_* = 26820$, fix $\alpha = 2117/500000$, $\beta = 209/100$, and put $D = e^{tL}$, $t = 1/2 - \alpha$, $H = e^{L/2}/L^\beta$. Set $L_0 = 24799$ and $c = \log(Q_* H_0)$. On the closed interval $L_0 \leq L \leq L_*$, the support condition $\alpha L - \beta \log L > c$ persists since $\alpha - \beta/L_0 > 0$. Also $H \geq 10^9$, $H \leq \sqrt{N}$, $T < z$, $3 < s < 4$ and $k > 1$ hold throughout. Using (29) in the pointwise GRH AP bound gives the normalized leading expense

$$\frac{3\nu_*}{2U_*} (a_* + b_T e^{-tL}) P(L) L^{3-\beta}.$$

Because $0 < \beta < 3$, this expression is not claimed to decrease on a whole half-line. Instead, use the constant upper bound

$$C_{\text{AP}} = \frac{3\nu_*}{2U_*}(a_* + b_T e^{-tL_0})P(L_0)L_*^{3-\beta} \quad (30)$$

throughout this finite interval. This is valid because $P(L)$ and e^{-tL} decrease, and $L^{3-\beta} \leq L_*^{3-\beta}$. The positive floor charge is kept; at L_0 , $b_T e^{-tL_0} < 10^{-5302}$.

Replace only the leading AP term by C_{AP} . Both other AP terms described after (24), every prime-modulus centering charge, (23), and the finite loss are retained. The bilinear estimate still uses the maximum modulus H . Its proof in Section 6 needs $H \leq \sqrt{N}$ and $H \geq 10^9$, both verified here; it does not use the smaller support count. The endpoint certificate gives $s = 3.966128$ and

Term	Certified value
M_0	> 7.806706554276225
M_1	< 6.498953398927134
M_2	< 1.306496415897830
E_{AP}	< 0.001235093952316
Support slack	> 0.001974440924078
Surplus	> 0.000021645498948
E_{rect}	$< 10^{-194}$
E_{fin}	$< 10^{-1337}$

For these fixed parameters M_0 is constant, while M_1 , M_2 , the other AP terms and all remaining expenses decrease for $L \geq L_0$. Thus the strict surplus exceeding 10^{-5} at L_0 holds throughout $[L_0, L_*]$. Section 6.2 already proves the same claim for every $L \geq L_*$, with common normalization $U_N N/L^2$. These inclusive ranges cover the whole half-line $L \geq L_0$ and prove the preceding starting point e^{24799} using GRH alone. The preceding stronger 0.0179 coefficient at $L \geq 31000$ retains its earlier half-line certificate.

The separate-formula check reproduces 12 endpoint quantities and regenerates the integer product with an independent prime sieve. The histogram checks enumerate 9216 subsets and verify 17172 exact rational coefficients at $\sigma = 0, 1$; the integer checks cover 16384 divisor-floor terms and 24316 roughness patterns. An additional finite enumeration checks 8608 prime-factor modulus patterns, including the injective aggregate map and its rough-factor support count. These finite checks accompany the two general proofs above. The 135 endpoint candidates are a finite parameter grid, not a global optimality proof. This analytic-regime join does not connect to the finite Goldbach verification endpoint $4 \cdot 10^{18}$.

6.4 The union of the accepted medium supports

The preceding logarithmic count includes medium products that neither Rosser sign accepts. The following positive count keeps every preceding acceptance condition. It bounds the union of the two supports once, without changing the sieve coefficients in Lemma 16.

Lemma 21 (Six-state union count). *Let $D_0 = T^u$, $u > 3$, and let $H_0 = \kappa D_0 > 1$ bound both pure-medium Rosser supports as in Lemma 15. Use the complete medium-prime set, the degree bound K , labels $b(p)$ and cutoff J of Lemma 19, with $0 < h < \log \min \mathcal{M}$. Process the complete blocks in decreasing label order. In block $\mathcal{M}_b$, put*

$$c_b = \left\lceil \frac{\log D_0 - 3 \log \min \mathcal{M}_b}{h} \right\rceil, \quad m_b = |\mathcal{M}_b|.$$

For a positive polynomial $F = \sum_B F_B v^B$, define the positive operators

$$\mathcal{L}_c F = \sum_{B < c} F_B v^B, \quad \mathcal{H}_c F = \sum_{B \geq c} F_B v^B, \quad \mathcal{S}_b F = v^b F \pmod{v^J}.$$

Track the vector $X = (U_e, U_o, L_e, L_o, B_e, B_o)^\top$. The first two coordinates mean only the upper sign is alive, the next two mean only the lower sign is alive, and the final two mean both signs are alive; subscripts record the parity of the number of selected primes. Start with $B_e = 1$ and every other coordinate zero. Within the block, the append operator is

$$\mathcal{A}_b X = \begin{pmatrix} \mathcal{S}_b(U_o + \mathcal{H}_{c_b} B_o) \\ \mathcal{S}_b \mathcal{L}_{c_b} U_e \\ \mathcal{S}_b \mathcal{L}_{c_b} L_o \\ \mathcal{S}_b(L_e + \mathcal{H}_{c_b} B_e) \\ \mathcal{S}_b \mathcal{L}_{c_b} B_o \\ \mathcal{S}_b \mathcal{L}_{c_b} B_e \end{pmatrix}. \quad (31)$$

Apply the complete-block update

$$X \longleftarrow \sum_{j=0}^{\min(K, m_b)} \binom{m_b}{j} \mathcal{A}_b^j X. \quad (32)$$

Then $A_{\text{acc}} := \sum_i X_i(1)$ bounds the number of distinct medium products accepted by at least one Rosser sign. The same complete-set bound holds after arbitrary prime deletion, retaining the complete K, J, c_b and replacing only m_b by the number of available primes.

Proof. At new selected length r , the upper sign checks odd r and the lower sign checks even r . If d_{old} is the preceding product, a checked append of prime p passes only if $d_{\text{old}} p^3 < D_0$. Its preceding label B has $hB \leq \log d_{\text{old}}$, so actual passing implies

$$B < \frac{\log D_0 - 3 \log p}{h} \leq \frac{\log D_0 - 3 \log \min \mathcal{M}_b}{h}, \quad B < c_b.$$

Consequently, a true surviving sign is never killed by the relaxed cutoff. Unchecked appends preserve that sign. For both signs alive, exactly one sign is checked at each append: a high-label append leaves the other sign alone. This gives (31). Each label and flag state has at most one destination; the low and high branches are disjoint. Induction shows that the relaxed alive set contains the true alive set, and every retained relaxed subset occurs in exactly one of the six exclusive states. Subsets with no surviving sign may be dropped permanently.

All primes in a block have the same label and relaxed cutoff. Choosing j distinct primes therefore applies the same operator j times, with exact multiplicity $\binom{m_b}{j}$. Their actual decreasing order does not affect this relaxed update. Every true final accepted product is below H_0 , hence contains at most K primes and has label below J . Every preceding label is no larger, so neither the block degree cutoff nor intermediate truncation can remove that true contribution. Excessive total degrees and possible passes retained by rounding are additional positive terms. The resulting sum bounds the union, without counting a subset twice when both signs survive.

After deletion, every surviving prime still belongs to its complete block. The same complete minimum supplies a valid necessary passing condition; the complete K, J still cover every accepted product. The operators have nonnegative coefficients and $\binom{m'_b}{j} \leq \binom{m_b}{j}$ for $m'_b \leq m_b$. Induction over the blocks therefore gives coefficientwise domination in every state. No subtraction of uncertain intervals is required. $\square$

Replace A_σ by A_{acc} in (29) and set

$$\nu_{\text{acc}} = \frac{2^{\omega(Q_*)} A_{\text{acc}}}{Q_* H_0}. \quad (33)$$

The composite coefficient support is bounded by the union of the two medium supports and a T -rough large factor, so the same proof gives $\nu_{\text{acc}}(a_T + b_T/D)H$ for both sieve signs. For the whole A_q sum, absorb q into the large factor as in Section 6.3. The medium product is still in this same union; the unique prime at least z makes $(q, d) \mapsto qd$ injective over the entire sum. Thus the new count also bounds its aggregate image. The integer roughness boundary, prime-modulus centering, and bilinear charges are unchanged.

For $T = 5000000$, $w_0 = 5$, $u = 87/16$, $\kappa = 11/169$ and count width $h = 1/100$, the complete count has $K = 22$, $J = 8115$ and 992 blocks. It gives

$$A_{\text{acc}} = 655257348020585286337311701158, \quad \nu_{\text{acc}} < 1.007455 \cdot 10^{-6} < 1/992601.$$

The preceding unfiltered A_0 is more than 8.37 times as large. The gap budgets still use their independently certified width $1/200$.

Coverage of all larger integers. For the lower starting point use $L_0 = 24274$, $L_* = 24799$, $\alpha = 1057/250000$ and $\beta = 93/50$; the medium family is $u = 87/16$. Equation (30), with the new $\nu_* := 1/992601$ and the same $a_* := 36411450583/10^{12}$, supplies a constant leading AP expense throughout $[L_0, L_*]$. The support condition $\alpha L - \beta \log L > \log(Q_* H_0)$, $H \geq 10^9$, $H \leq \sqrt{N}$, $T < z$, $3 < s < 4$ and $k > 1$ hold at L_0 and persist. In particular $\alpha - \beta/L_0 > 0$ and $1/2 - \beta/L_0 > 0$. The finite charge $b_T e^{-tL_0}$ is positive and below 10^{-5189} ; it has not been discarded. The fixed-parameter monotonicity argument in Section 6.3 proves the required surplus over this entire closed interval. The preceding interval $[24799, 26820]$ and the preceding half-line $[26820, \infty)$ use the same 10^{-5} claim and normalization. Together these inclusive ranges prove the first assertion of Theorem 1 for every $L \geq 24274$.

For the stronger claim, use $u = 11/2$, the same T, w_0, κ , $\alpha = 3737/1000000$, $\beta = 3$ and $L_0 = 31000$. Its gap budgets are $\eta_+ = 1/3231$, $\eta_- = 1/3242$, with $\epsilon = 1/41219$. The accepted count gives $\nu_{\text{acc}} < 8.498637 \cdot 10^{-7} < \nu_* := 1/1176659$, and $a_T < a_* := 36409264976/10^{12}$. Now $L^{3-\beta} = 1$, so the leading AP expense

$$\frac{3\nu_*}{2U_*}(a_* + b_T e^{-tL})P(L)$$

decreases on the whole half-line. Its value at 31000 is a valid constant upper bound for every $L \geq 31000$. All other losses decrease and the support slack increases. This separate half-line argument proves the stronger coefficient 0.0181 of Theorem 1; it does not transfer a larger claim through an earlier certificate with a smaller surplus. The bilinear estimate continues to use the maximum modulus H .

Term	$L_0 = 24274$	$L_0 = 31000$
α	1057/250000	3737/1000000
β	93/50	3
s	3.966176	3.970104
M_0	> 7.806728439046083	> 7.809030620429977
M_1	< 6.498811372057018	< 6.485781769136601
M_2	< 1.306485180691536	< 1.305068915489237
E_{AP}	< 0.001407561467773	< 0.000000011604981
Support slack	> 0.000548291572001	> 0.008509203429583
Surplus	> 0.000024324829757	> 0.018179924199159
E_{rect}	$< 10^{-188}$	$< 10^{-250}$
E_{fin}	$< 10^{-1300}$	$< 10^{-1673}$

The separate-formula program reproduces 12 quantities for each new endpoint, regenerates all 348513 integer primes through T , and rechecks the earlier interval and tails. Exact small-set checks cover 4608 complete subsets, 114876 integer histogram coefficients, 104976 deleted-subset patterns and 384 deleted production histograms. The earlier 8608 aggregate prime-factor patterns remain applicable. The 70 weak-endpoint candidates and nine stronger-count candidates are a finite search, not a proof of global optimality. These checks supplement the general proof above; they do not replace the external analytic inputs or independent research review. No distribution hypothesis beyond GRH has been introduced, and this analytic interval join still does not reach finite verification.

7 A parameterized EH-style reduction

For a quantitative comparison with standard distribution conjectures, put

$$\mathcal{E}_\theta(x) = \sum_{d \leq x^\theta} \mu^2(d) \max_{(a,d)=1} |\pi(x; d, a) - \pi(x)/\varphi(d)|.$$

Lemma 22 (Deleted-prime remainders). *If $10^9 \leq H \leq N^\theta$, then, with the density $g(d) = 1/\varphi(d)$,*

$$\sum_{\substack{d \leq H \\ d|P_N(z)}} |r_A(d)| \leq \mathcal{E}_\theta(N) + \omega(N)\{2\sqrt{N} + 1.1 \log H\}.$$

Proof. Let κ be the number of deleted primes dividing N , and let κ_d count those with $d \mid N - p$. Since $(d, N) = 1$,

$$r_A(d) = \pi(N; d, N) - \frac{\pi(N)}{\varphi(d)} - \kappa_d + \frac{\kappa}{\varphi(d)}.$$

Each $N - p < N$ has at most $2\sqrt{N}$ positive divisors, and $\kappa \leq \omega(N)$. Summing and using (15) proves the claim. $\square$

The usual EH conjecture provides asymptotic estimates with parameter dependent implied constants; see [12, Claim 2.2]. It does not prescribe a numerical constant or a numerical starting point. We retain these two quantities rather than assign them convenient values.

Theorem 23 (Quantitative distribution inputs). *Suppose a finite distribution input provides numbers $C > 0$ and X such that*

$$\mathcal{E}_\theta(x) \leq Cx/\log^4 x \quad (x \geq X). \quad (34)$$

No RH or GRH is assumed. The following conclusions hold:

θ	threshold for even N	$\pi_2(N)L^2/N$
3/4	$\max\{X, \exp(\max\{786, \sqrt{20C}\})\}$	> 0.01
9/10	$\max\{X, \exp(\max\{212, \sqrt{10C}\})\}$	> 0.02
19/20	$\max\{X, \exp(\max\{166, \sqrt{10C}\})\}$	> 0.02
99/100	$\max\{X, \exp(\max\{142, \sqrt{10C}\})\}$	> 0.02

No numerical value of C or X is asserted by this theorem. There are also weaker counting bounds $\pi_2(N) > 10^{-5}N/L^2$ with the same square-root constraints and the smaller fixed starts 774, 210, 165, and 141, respectively.

Proof. Use the finite Rosser medium weights of Lemma 9, with $D_0 = T^u$, and Lemmas 11 and 13 with $M = 10$, $\Sigma_E = \{0\} \cup \{a/20 : 2 \leq a \leq 12\}$, and $\Gamma_E = \{0, 3/25, 6/25, 9/25\}$. Take the following fixed parameters:

θ	w_0	T	u	M	ϵ^{-1}	η_+^{-1}	η_-^{-1}	(C_1, C_2)
3/4	3	22000	39/8	10	1717	141	143	(113, 114)
9/10	3	10000	35/8	10	1012	46	45	(113, 114)
19/20	3	10000	17/4	10	1012	34	33	(113, 114)
99/100	3	10000	17/4	10	1012	34	33	(113, 114)

For $\theta = 3/4$ Lemma 8 gives $\widehat{K}_T - 1 < 0.000582284267$ and separate Rosser gaps $\widehat{R}_+^{(\text{band})} < 0.007043548708$, $\widehat{R}_-^{(\text{band})} < 0.006945935768$. For $\theta = 9/10$ the ordered product error is less than 0.000987573450, and the gaps obey $\widehat{R}_+^{(\text{band})} < 0.021591509791$, $\widehat{R}_-^{(\text{band})} < 0.021953570359$. The higher two levels use the same product error and gap bounds, less than 0.029170196992 and 0.029947327150, respectively. The finite sums use all 2462 and 1227 medium primes, respectively. They are strictly below the chosen ϵ, η_+, η_- . Take $z = N^{1/3}$, $H = N^\theta$, and $D = H/(\kappa Q_* D_0)$, using Lemma 15. Put $c = \log(\kappa Q_*) + u \log T$. Then $s = 3\theta - 3c/L$ and $\kappa Q_* D_0 D = H$. Here $Q_* = 3$ and $\kappa = 5/49$ throughout; the costs are $c < 47.561$, $c < 39.112$, and $c < 37.961$ for the two higher levels. At all four stated starting points, $s > 2$, so $D \geq z^2$. The refined inequalities (8) apply with the uniform product condition and Lemmas 10, 11, and 13. Using JS Lemma 3.2 and the exact product identity gives, unconditionally,

$$|A|V(z) \geq \frac{3U_N N}{L^2} \rho_U(L), \quad \rho_U(L) = 1 - \frac{20.508 \cdot 10^{-6}}{L} - 2Le^{-L/3}.$$

Indeed $z > 10^{12}$ throughout, $D(z)/D_\infty \geq \exp(-6.836 \cdot 10^{-6}/\log z)$, and the missing factors for $p \mid N$ cost at most $2L/z$. The count $|A| > N/L$ is unconditional by JS Lemma 3.1. The normalized lower main coefficient is at least

$$3U_* \rho_U(L) \{(1 + \eta_+)(f(s) - C_2 \epsilon e^{2-s}) - (\eta_+ + \eta_-)(2e^\gamma/s + C_1 \epsilon e^{2-s})\}.$$

For $2 < s < 3$, $f(s)$ increases while both subtracted corrections decrease. Thus this coefficient increases with L at every listed level. The input costs C/L^2 , which is at most $1/20$ or $1/10$ under the corresponding square-root constraint. Lemma 22, with $\omega(N) \leq L/\log 2$, costs at most

$$\frac{L^3}{\log 2} \{2e^{-L/2} + 1.1\theta L e^{-L}\},$$

and excluding the survivor one costs $L^2 e^{-L}$. Both decrease on these ranges. The certificates give main coefficients greater than 0.0607006692223 at $(\theta, L) = (3/4, 786)$ and 0.1228594687806 at $(9/10, 212)$. At the higher levels they exceed 0.1220961708146 at $(19/20, 166)$ and 0.1354387463886 at $(99/100, 142)$.

After all charges the surpluses exceed 0.0107006692223, 0.0228594687806, 0.0220961708146, and 0.0354387463886, respectively. For the weaker counting bounds keep every finite parameter and error budget fixed, and evaluate the same expression at $L = 774, 210, 165, 141$. The resulting full surpluses exceed 0.0003941826699, 0.0050969071192, 0.0083045836547, and 0.0166813945298, respectively. The same support, product, $2 < s < 3$, deleted-prime and survivor-one checks hold at these starts. Their monotonicity extends each bound to its whole half-line. An element $N - p$ of A is coprime to N , so it has no prime factor omitted from $P_N(z)$. Every remaining survivor other than one therefore has all prime factors at least z , and at most two prime factors, because it is less than $z^3 = N$. This proves all four conclusions. $\square$

More generally, under (34) at the specified level, the same argument gives a usable criterion for any fixed $2/3 < \theta < 1$. Choose valid parameters of Lemma 16, let $c = \log(Q_* H_0)$, and put $s = 3\theta - 3c/L$. For $N \geq X$, $L > 3\log(10^{12})$, $T < N^{1/3}$, and $2 < s < 3$, define

$$\begin{aligned} \Xi(L) = & 3U_* \rho_U(L) \{f(s) - C_2 \epsilon e^{2-s} - \eta(2e^\gamma/s + C_1 \epsilon)\} - \frac{C}{L^2} \\ & - \frac{L^3}{\log 2} \{2e^{-L/2} + 1.1\theta L e^{-L}\} - L^2 e^{-L}. \end{aligned} \quad (35)$$

Whenever $\Xi(L) > 0$, the sieve factor is positive and the preceding proof gives $\pi_2(N)L^2/N \geq \Xi(L) > 0$. All parameter and distribution conditions must be supplied; the criterion introduces no numerical value for C or X . Since $s < 3$, its negative corrections decrease as L increases for fixed parameters, and the positive main factor increases. Thus a positive starting-point bound extends to the corresponding half-line. With separate gap bounds one can replace the sieve factor by $(1 + \eta_+)l(s) - (\eta_+ + \eta_-)u(s)$, using $l(s) = f(s) - C_2 \epsilon e^{2-s}$ and $u(s) = 2e^\gamma/s + C_1 \epsilon e^{2-s}$. This factor is also increasing for $2 < s < 3$; the four rows above use this refined version.

The existence of unspecified constants of the type in (34) follows from the usual EH conjecture. One may transfer its von Mangoldt formulation to prime counting with a small loss in distribution level and logarithmic power, using partial summation; ordinary EH supplies all such fixed levels below one and all logarithmic powers. This observation does not make the constants effective or supply $X \leq e^{212}$. For instance, substituting $C = 1$ and $X = e^{212}$ in the second row would be a new finite hypothesis. We do not make that substitution or report e^{212} as a numerical threshold or record under ordinary EH alone.

8 Signed remainders and the support actually used

The absolute modulus sum is sufficient but is more than the lower sieve uses. Fix one of the four parameter sets in Theorem 23. At each even N in the sieve domain use its composite lower coefficients λ_N^- , with $z = N^{1/3}$ and $D = N^\theta/(Q_* H_0)$. Define

$$G_N^- = \sum_d \frac{\lambda_N^-(d)}{\varphi(d)}, \quad \mathcal{R}_N^- = \sum_d \lambda_N^-(d) \left(\pi(N; d, N) - \frac{\pi(N)}{\varphi(d)} \right).$$

These sums have precisely the support of the lower weights, contained in $d \mid P_N(z)$, $d < N^\theta$.

Lemma 24 (Signed lower remainder and deletions). *For the sequence $A = \{N - p : p < N, p \text{ prime}, p \nmid N\}$,*

$$S(A, z) \geq \pi(N)G_N^- + \mathcal{R}_N^- - \omega(N).$$

Proof. First sieve $A^* = \{N - p : p \leq N, p \text{ prime}\}$. Since N is even and at least four, no term has $p = N$. Summing the pointwise lower divisor inequality of Lemma 16 over A^* gives

$$S(A^*, z) \geq \sum_d \lambda_N^-(d) \pi(N; d, N) = \pi(N)G_N^- + \mathcal{R}_N^-.$$

Deleting the primes dividing N removes at most $\omega(N)$ survivors. This proves the claim without taking absolute values of their divisor sums. In particular, the $2\sqrt{N}$ deletion cost in Lemma 22 is unnecessary when only this signed lower remainder is used. $\square$

The following is a *tailored quantitative input*, not a newly named standard conjecture. Let $\mathfrak{D}_\theta$ be the even integers in the preceding domain $L > 3 \log(10^{12})$, $T < z$, $2 < s < 3$. For the specified family of weights, suppose

$$\mathcal{R}_N^- \geq -\frac{CN}{\log^4 N} \quad (N \geq X, N \in \mathfrak{D}_\theta). \quad (36)$$

The complete input (34) implies (36), with the same C, X , because $|\lambda_N^-(d)| \leq 1$. Only one side of one signed sum is required here. Neither GRH nor a published weighted asymptotic estimate supplies its numerical C, X . The integrality lower bound for the complete maximum-error sum does not, by itself, give a lower bound for this signed sum.

Put $c = \log(Q_* H_0)$, $s = 3\theta - 3c/L$, and use the separate gaps of Theorem 23. With its functions $l(s), u(s)$, write

$$M_\theta(L) = 3U_* \rho_U(L) \{(1 + \eta_+)l(s) - (\eta_+ + \eta_-)u(s)\}.$$

Lemma 24, the same product and prime-count inputs, and $\omega(N) \leq L/\log 2$ give the sharper criterion

$$\pi_2(N) \frac{L^2}{N} \geq \Xi_w(L) := M_\theta(L) - \frac{C}{L^2} - \left(\frac{L^3}{\log 2} + L^2 \right) e^{-L}, \quad (37)$$

when the main factor is positive and the preceding sieve domains hold. The last $L^2 e^{-L}$ charge removes the possible survivor one. For a requested surplus $a > 0$, the admissible budget at L is therefore

$$C < L^2 \left\{ M_\theta(L) - \left(\frac{L^3}{\log 2} + L^2 \right) e^{-L} - a \right\}.$$

This spends the actual positive main mass, instead of imposing the fixed 1/20 or 1/10 budgets of Theorem 23.

Proposition 25 (Finite-constant budget conversion). *Suppose (36) holds for one of the four specified weight families. Each entry A below gives $\pi_2(N) > 10^{-5}N/\log^2 N$ for every even $N \geq \max\{X, e^A\}$, provided C does not exceed the indicated column label:*

θ	$C \leq 1$	$C \leq 10^3$	$C \leq 10^4$	$C \leq 10^6$
3/4	722	723	740	1632
9/10	200	202	222	936
19/20	158	161	182	872
99/100	136	138	159	829

The same conclusions hold under the complete input (34). The column labels are allowed budgets, not distribution constants proved for the primes. The onset X is retained in every conclusion.

Proof. The 192-bit certificate evaluates (37) strictly above 10^{-5} at each entry, checks all sieve domains, and reproduces the saved main masses of Theorem 23 within 10^{-45} . It also evaluates the main mass by an independent algebraic expression in which the e^γ factors are cancelled explicitly. On $2 < s < 3$, $l(s)$ increases and $u(s)$ decreases, so the composite factor increases. The positive $\rho_U(L)$ increases on these ranges. Both exponential charges and C/L^2 decrease. Hence the inequality extends to the whole half-line. For smaller C the bound only improves. The P_2 transfer is the same as in Theorem 23. $\square$

For example, if a quantitative input is supplied with upper constant $C = 10^6$ at level 99/100, the earlier square-root allocation requires $L \geq \sqrt{10^7} = 3162.277\dots$, whereas Proposition 25 requires $L \geq 829$. Both retain the same unknown onset X . This is an improvement of the budget conversion under the same input; it is not a numerical record under ordinary EH. Even with zero charged remainder the current certified expression has negative main factor at $L = 721, 199, 157, 135$ for the four rows. Its first positive integer starts are 722, 200, 158, 136. This limitation concerns these fixed weights and main-mass bounds.

8.1 Prefix-block finite-constant conversion

Recompute the finite medium masses using Lemma 14. The following five families use $w_0 = 3$, $Q_* = 3$, $\kappa = 5/49$, $D_0 = T^u$, and the indicated gap and product budgets:

Family	T	u	h	ϵ	η_+	η_-	(C_1, C_2)
1	22000	33/8	1/200	1/1717	1/100	1/103	(113, 114)
2	100000	9/2	1/100	1/4729	1/233	1/235	(109, 110)
3	10000	29/8	1/200	1/1012	1/32	1/31	(113, 114)
4	10000	15/4	1/200	1/1012	1/42	1/42	(113, 114)
5	10000	7/2	1/200	1/1012	1/24	1/23	(113, 114)

For a chosen cell, use that family's composite lower weights at $z = N^{1/3}$ and $D = N^\theta/(\kappa Q_* D_0)$. If only the signed input (36) is assumed, it must hold for this specified family on its sieve domain. The full absolute input (34) implies it for every family with the same C, X . No value of C or X is supplied by this construction.

Proposition 26 (Prefix-block budget conversion). *Under that cell's signed input, or the full absolute input, every even $N \geq \max\{X, e^A\}$ satisfies $\pi_2(N) > 10^{-5}N/\log^2 N$. Here each entry is A followed by its family number, and C is bounded by the column label:*

θ	$C \leq 1$	$C \leq 10^3$	$C \leq 10^4$	$C \leq 10^6$
3/4	627 (1)	629 (1)	648 (1)	1510 (2)
9/10	171 (3)	174 (3)	198 (4)	885 (2)
19/20	135 (5)	138 (5)	162 (3)	825 (2)
99/100	115 (5)	118 (5)	143 (3)	791 (2)

Proof. The complete-set bounds of Lemma 14 supply the two gaps in (8); all coefficient, support and signed remainder identities continue to apply. Use exactly (37) with each family's $c = \log(\kappa Q_*) + u \log T$. The 192-bit certificates check every cell's strict surplus, $T < z$, $L > 3 \log(10^{12})$, and $2 < s = 3\theta - 3c/L < 3$. For fixed parameters, $l(s)$ increases and $u(s)$ decreases on $(2, 3)$, s and $\rho_U(L)$ increase, and every deducted expense decreases. Thus every cell extends to the claimed half-line. The exact and separate-formula checks test finite arithmetic, not a distribution upper bound or independent mathematical review. $\square$

Ordinary EH still supplies only unspecified constants and starting points. These entries improve the conversion of a supplied quantitative input, and are not numeric EH-only records. In particular X cannot be discarded. The smallest support cost among these five families is 31.052421204; even $\theta < 1$ allows only $L_B/3 < 14.278$ at B . Thus these families still cannot reach the verified endpoint, even with zero signed error. The complete absolute input also retains the integrality constraints of Proposition 4.

Proposition 27 (A support obstruction to triple factorability). *The present four lower-weight families are not triply well-factorable of level N^θ on a whole half-line.*

Proof. For a prime $T < p < z$ with $p \nmid N$, the lower Rosser weight has no checked even prefix, and the empty medium gap is zero. Consequently $\lambda_N^-(p) = -1$. A triply well-factorable sequence of level H must have coefficient zero at every prime $p > H^{1/3}$: apply the balanced split $H_1 = H_2 = H_3 = H^{1/3}$ in the definition [1, Definition 1.1]. Every factorization of a prime has a factor equal to p , outside its allowed support.

For the four rational levels, take $N = 2^{3000k}$, $k \geq 1$. Then $H^{1/3} = 2^{1000k}$ is an integer. Bertrand's theorem provides a prime strictly between $H^{1/3}$ and $2H^{1/3} < z$. It exceeds T and cannot divide N . The parameter certificate checks $D \geq z^2$ already at $k = 1$, and that support slack increases with k . This prime has coefficient -1 , a contradiction. The certificate also supplies the exact small witness $N = 10^{18}$, $z = 10^6$, $T = 5$, $D_0 = 625$, $D = 10^{14}$: $p = 999983$ is prime, its coefficient is -1 , and $p^3 > H = 3D_0D = 187500000000000000$. $\square$

This does not decide two-factor well-factorability or the applicability of a separately constructed well-factorable linear-sieve variant. Changing the coefficient family requires a new pointwise and main-mass proof; support size alone is not enough.

Pascadi's fixed-residue theorem gives $5/8 - \varepsilon$ for triply well-factorable weights *unconditionally*, and $3/5 - \varepsilon$ for its upper linear-sieve variant; Theorem 5.4 treats specified lower blocks with factor-dependent levels [1]. Its constants depend on the residue a , so setting $a = N$ does not give the required uniform input. Lichtman's large-residue formula is $(5 - \vartheta_{\text{RP}})/8 - \varepsilon$, giving $153/256 - \varepsilon$ at $\vartheta_{\text{RP}} = 7/32$ [2, Theorem 1.7]. Setting $\vartheta_{\text{RP}} = 0$ requires the Ramanujan–Petersson input identified there, rather than Dirichlet GRH alone. Even the resulting $5/8$ is below the $2/3$ needed for this direct $z = N^{1/3}$ linear lower sieve; Chen switching remains necessary for such lower distribution levels.

The MPZ smooth-modulus input allows a primitive residue depending on N , but requires prime factors of its moduli at most N^δ . The proved region in [12, Claim 2.4, Theorem 2.5] is $600\varpi + 180\delta < 7$ at level $1/2 + 2\varpi$. The current nonzero prime coefficients extend to $N^{1/3}$ on a half-line, so their uniform coverage needs $\delta \geq 1/3$: apply Bertrand near z along $N = 2^{3000k}$. This is outside that region, whose level supremum is only $157/300 < 2/3$. A restriction to smooth moduli would leave uncontrolled coefficients and needs a different decomposition.

Li's 2026 theorem states the asymptotic lower count $1.9728C_N N / \log^2 N$, combining weighted sieves, switching, the double sieve and Harman's sieve [3]. Its statement does not supply a numerical starting point. Explicit uniform estimates for its switched blocks are a route to investigate, not an explicit threshold certified here.

9 Cost calculation for the rejected bridge candidate

Withdrawal notice. Proposition 3 refutes the distribution premise of this section. The following algebra is retained to identify the original error: it checks that a proposed budget would suffice, without checking whether that budget is possible. It does not prove a nonvacuous full-range result.

Here we do not use the GRH rectangle machinery. Put $z = N^{1/3}$, $H = N^{9/10}$, $D = H/e^8$, $N \geq B$. Lemma 5 permits $\epsilon = 1/700$, $C_2 = 117$, and

$$s(L) = 2.7 - 24/L \geq s(L_B) = 2.139682079805 \dots > 2.$$

The former calculation had $D \geq z^2$ and $Q_N D \leq H$; its fixed-endpoint-only sieve application was not independently justified. For $2 \leq s \leq 2.7$, $f(s)$ increases while $e^2 h(s) = e^{2-s}$ decreases. Hence the positive sieve factor is at least

$$f_B = f(s(L_B)) - (117/700)e^{2-s(L_B)} > 0.0723184176500.$$

Also (10) and (11) give

$$|A|V(z) \geq \frac{3U_N N}{L^2} \rho(L), \quad \rho(L) = 1 - \frac{L}{8\pi} e^{-L/6} - 2Le^{-L/3}.$$

The function ρ increases on this range, and $\rho(L_B) > 0.9985933587896$.

QEH_B concerns all primes, while A excludes $p \mid N$. We explicitly account for the difference. Write $\kappa = \#\{p < N : p \mid N\}$ and $\kappa_d = \#\{p < N : p \mid N, d \mid N - p\}$. For $(d, N) = 1$,

$$r_A(d) = \pi(N; d, N) - \frac{\pi(N)}{\varphi(d)} - \kappa_d + \frac{\kappa}{\varphi(d)}.$$

For each deleted prime p , the integer $N - p$ has at most $2\sqrt{N}$ positive divisors. Therefore, irrespective of H ,

$$\sum_{\substack{d \leq H \\ d \mid P(z)}} |r_A(d)| \leq \frac{N}{10L^2} + \omega(N)\{2\sqrt{N} + 1.1 \log H\}. \quad (38)$$

This divisor count avoids the wasteful $\omega(N)H$ estimate. Use Robin's bound $\omega(N) < 1.3841L/\log L$, recorded in [5]. After normalizing by N/L^2 and excluding the possible survivor one,

$$\begin{aligned} \pi_2(N) \frac{L^2}{N} &\geq 3e^{-\gamma} \rho(L_B) f_B - \frac{1}{10} \\ &\quad - \frac{1.3841L^3}{\log L} \left(2e^{-L/2} + 0.99Le^{-L} \right) - L^2 e^{-L}. \end{aligned} \quad (39)$$

An element of A surviving $P_N(z)$ is coprime to N , hence has no excluded prime factor. Its prime factors are all at least z . Three factors would give an integer at least $z^3 = N$, whereas every element of A is less than N . So every surviving element other than one is a legitimate prime or semiprime. The error terms on the second line of (39) decrease for $L \geq L_B$. At B the main coefficient exceeds 0.1216402390842; the deleted-prime budget is less than 0.0000289481782 and $L_B^2/B < 4.59 \cdot 10^{-16}$. The final surplus exceeds 0.0216112909060 > 0.021 . The computation of [11] covers every even $4 \leq N \leq B$ by two primes. Nevertheless the proposed analytic premise fails at B , so this calculation does not establish a bridge.

10 Interpretation and remaining research questions

The GRH theorem changes finite constants and presieving cost, while retaining the asymptotic square-root distribution level. The composite weights resolve the endpoint problem for this theorem, but its threshold still does not bridge the verified range. Under the usual GRH absolute-error

summation, direct sieving at $z = N^{1/3}$ has asymptotic parameter $s \leq 3/2$; the lower linear sieve has no positive main term there. Chen’s switching avoids that obstruction at large N , with a finite support and error cost that remains large near B . For the present direct distribution route, $D = N^\theta/e^c$ and $z = N^{1/3}$ require $c \leq (\theta - 2/3)L$ just to have $D \geq z^2$. At $B = 4 \cdot 10^{18}$, $L_B = 42.832826035\dots$, the allowances are only 3.569402170 and 9.994326075 at the two chosen levels, whereas the earlier fixed-family costs are 47.560368848... and 39.111469030.... Thus even zero AP error would not make these fixed weights reach B . A substantially smaller support cost or a different sieve construction is needed for this route. This is an obstruction to the present construction, not an impossibility theorem for conditional Chen bounds.

The rejected QEH_B candidate changes the level and supplies an explicit constant and starting point. Its sieve budget would have been sufficient, but its distribution budget is impossible at B . Any replacement must first pass the integrality obstruction and then be related to a recognized conjecture or independently supported finite distribution input. Whole-range stitching remains open in this project. RH alone controls complete prime products and global prime counts, and has not supplied the AP input required by Theorem 1. Finite-height GRH verification cannot be substituted for full GRH without a certified explicit-formula tail.

For whole-range comparisons the 2026 thesis summary of Johnston [10] states GRH $p + P_{16}$, improving the older $p + P_{31}$. It must not be confused with a whole-range $p + P_2$ theorem. The withdrawn candidate supplies no improvement on this whole-range result. EH and GEH are recognized conjectures used for conditional prime-gap results [12]. GRH is not known to imply either. Their usual forms do not provide the concrete constants and starting points needed to connect a finite verification with an explicit analytic threshold. Theorem 23 records this dependence instead of hiding it.

Proposition 4 also constrains the optimization itself: the level 99/100 route of Theorem 23 has an actual threshold greater than e^{834} under that theorem’s square-root budget. Merely relaxing that budget cannot recover an endpoint near e^{141} within the present direct-sieve expression. Indeed, for (35) and its separate-gap version, assume the proved conditions $0 < \rho_U(L) \leq 1$, $2 < s \leq 3\theta < 3$, nonnegative support cost, and nonnegative error parameters. Since $f(s) < F(s)$ and f increases on this interval, the main coefficient is at most

$$3U_*f(3\theta) = \frac{7.92 \log(3\theta - 1)}{3\theta} =: K_\theta.$$

All other charges are nonnegative, so $\Xi(R) > 0$ requires $C < K_\theta R^2$. At $\theta = 99/100$, the strict certificate gives $\mathfrak{I}_\theta(492) > K_\theta 492^2$. If such a positive endpoint satisfied $R \leq 492$ and $e^R \geq X$, the input at $x = e^{492}$ would imply $C > \mathfrak{I}_\theta(492) > K_\theta 492^2 \geq K_\theta R^2$, a contradiction. Thus even after reducing support cost and choosing a larger error budget, this direct-sieve expression needs $R > 492$. This conclusion uses the stated normalization and main-term model; it is not a bound for all conditional Chen arguments.

Reproducibility and dependency ledger

For the accepted-support union and current endpoints, run:

```
conditional_chen_accepted_support_checks.py
conditional_chen_accepted_interval_certificate.py
conditional_chen_accepted_interval_checks.py
```

The counter uses six positive exclusive flag/parity states and exact binomial block multiplicities. Its complete cutoffs remain fixed after deletion. Counts are generated completely or reused only

after generator hash checks. The scalar checker uses its own endpoint formulas and integer prime sieve, checks exact sums of the six state counts, and rechecks the preceding interval and tails. The weak interval covers $[24274, 24799]$ and joins the earlier $[24799, 26820]$ and GRH tail; the stronger coefficient uses its own fixed- $\beta = 3$ half-line. Continuous coverage follows from Section 6.4, not from finitely many sample points. Neither a finite-verification splice nor independent research review is supplied by these programs. For the logarithmic count and rough-integer interval, run:

```
conditional_chen_binned_support_checks.py
conditional_chen_rough_interval_certificate.py
conditional_chen_rough_interval_checks.py
```

The support histograms are generated completely or reused from per-case caches after checking generator hashes. The interval program computes the complete integer Euler product freshly; the separate checker regenerates its prime set independently. It rechecks the old tail, verifies the common claim and inclusive join, and uses its own formulas for 12 new endpoint quantities. Continuous coverage follows from the fixed-parameter argument in Section 6.3, not from the five sample points. This work does not supply independent research review or a splice to the finite verification range. For the current prefix-block results, run the following files in order:

```
conditional_chen_prefix_block_checks.py
conditional_chen_prefix_block_endpoint_certificate.py
conditional_chen_prefix_block_refinement.py
```

The refinement records per-case caches tied to generator hashes. The initial grid includes a previously generated $u = 11/2$ finite sum, explicitly reused by hash. The separate support-count program `conditional_chen_sparse_modulus_certificate.py` freshly evaluates the complete Euler products at four rational exponents for each of three cutoffs; it also checks 18 exact small Rankin counts. Then run `conditional_chen_prefix_block_endpoint_checks.py`. It independently reproduces eight GRH quantities at each asserted endpoint and the main mass and surplus in all 16 distribution cells, without calling the production scalar evaluators. The exact prefix-block checks cover 4608 deleted prime subsets, 9216 first-failure identities, 104976 divisor patterns, and 132 block-coefficient groups. These tests do not replace Lemmas 14 and 18, the external inputs, independent research review, or a verification-range splice.

First run `python conditional_chen_composite_certificate.py` to regenerate the finite inputs and the conservative-support baseline. Then run `python conditional_chen_sharp_endpoint_certificate.py` for the preceding GRH endpoints and the distribution theorem. For the preceding dense-grid GRH endpoints, then run `python conditional_chen_dense_band_grh_certificate.py`. It freshly generates the enlarged-grid finite sums, saves `conditional_chen_dense_band_grh_finite.json`, and checks the new AP budget and endpoints. The optional argument `--reuse-finite` reuses that file after checking its generator hashes. Five quantities at the earlier $L = 30824$ are reproduced within 10^{-45} ; direct AP, surplus and support formulas are also checked within 10^{-45} . The ordered product input is reused with its hash, not regenerated in this additional run. The sharp endpoint program records the SHA-256 hashes of the saved finite inputs, scans the complete medium-prime sets again for the support maxima, and checks all new support, parameter and surplus conditions. Its independent scalar implementation also reproduces eight baseline GRH quantities within 10^{-45} before recharging the support. The first program regenerates the complete ordered-product certificate through 10^8 by calling `conditional_chen_uniform_product_certificate.py`; saved JSON values are not used as proof inputs in that first program. The independent finite

`conditional_chen_uniform_product_checks.py` checks 4096 integer sequences against exhaustive ordered pairs, 1954 exact prime product endpoint pairs, and nine integer/prime tail comparisons. Those checks exercise the merging and endpoint conventions, and do not replace Lemma 8 or its published tail inputs. The module `conditional_chen_rankin_certificate.py` generates all medium primes by a complete integer sieve and sums the positive Rankin bound in 192-bit Arb. The earlier `conditional_chen_degreewise_rankin_certificate.py` uses its integer sieve and degree thresholds, and maintains the positive recurrence separately for each exponent. The earlier `conditional_chen_quadratic_rankin_certificate.py` adds the moments at $0, \sigma, 2\sigma$, checks the optimizer signs strictly, and encloses the nonnegative quadratic expression. It is retained as a baseline and can also be run separately. The separate run writes a `_standalone.json` file, preserving the production JSON with its freshly computed uniform-product metadata. The separate `conditional_chen_degreewise_rankin_checks.py` compares nine finite coefficient configurations with exact rational subset sums and verifies 2304 first-failure and prime-deletion cases. The independent `conditional_chen_quadratic_rankin_checks.py` also checks nine configurations and 2304 cases, evaluating the optimized polynomial by exact rational subset sums and comparing actual first failures for all deleted prime subsets. The current `conditional_chen_prefix_band_certificate.py` maintains positive raw exponential-generating moments through degree four, encloses the signed shift and Gram optimization by Arb, and skips any optimizer whose signs cannot be certified. Native truncated polynomial multiplication computes the required coefficients exactly, without omitting any subset mass. Its separate run preserves the production JSON. The independent `conditional_chen_prefix_band_checks.py` uses formal power-of-two atoms, not primes, so that base-two logarithms and polynomial expectations are exact rational numbers. It checks twelve configurations, 2688 first-failure/deletion cases, and 25 actual product-band instances, including one two-atom first failure. These checks validate the finite algebra; they do not replace the general prime-set proof. The `conditional_chen_prefix_band_eh_search.py` explores 96 strict scalar candidates. Its four selected finite inputs are regenerated by the composite certificate; the sharp endpoint program then selects the new starts with the smaller proved support costs. The earlier quadratic search of 80 and degreewise search of 144 candidates remain baselines. The `conditional_chen_sharp_support_certificate.py` computes the exact support maxima by complete adjacent-prime scans and checks 52488 exact coefficient/deletion cases. The older `conditional_chen_product_certificate.py` and `conditional_chen_interval_certificate.py` are retained for the unused fixed-endpoint candidates and withdrawn bridge costs. The separate `conditional_chen_integrality_certificate.py` refutes both finite EH-style candidates. Only strict Arb comparisons accept a claim. The `conditional_chen_distribution_tradeoff_certificate.py` certifies the necessary constant/onset witnesses in Proposition 4 and the direct-sieve positivity obstruction above. It also computes actual reduced-class prime histograms for 19701 integer pairs $3 \leq x \leq 200$, $2 \leq H < x$, checking the prime-modulus subtotal against the integrality lower bound with exact rational counts. These finite checks do not replace the general proof or provide any distribution upper bound. The `conditional_chen_weighted_distribution_certificate.py` reproduces the saved distribution main masses and certifies Proposition 25. It uses exact rational arithmetic in 10748 composite cases over 1156 selected even integers, checks 99158 divisor patterns and 21496 signed identities, including deleted primes and the P_2 transfer. It verifies the finite prime witness in Proposition 27 and the support inequalities for its infinite families. It does not prove a signed distribution upper bound, regenerate the large finite prime products, or certify global optimality. The exact-integer script `conditional_chen_weight_checks.py` also checks 2347008 pointwise divisor patterns across six finite exact/Brun/Rosser configurations, including the integer levels on either side of every weight transition. These finite checks exercise the coefficient, support and point-

wise inequalities; they do not replace the general proof of Lemma 16 or the external main-mass estimates. The `conditional_chen_rosser_checks.py` checks 602368 pointwise patterns across three finite exact/Rosser/Rosser configurations, eight additional rational-power configurations, 16 normalized coefficient cases, and 3088 exact first-failure and prime-deletion cases. The integer configurations include both sides of every weight transition; the rational-power ones compare levels by exact integer powers, without numerical roots. They also check the separate parity bounds by explicit subset sums and the minimum-prime degree restrictions by integer powers, including the eight rational powers 5, 9/2, 35/8, 25/4, 17/4, 49/8, 39/8, and 33/8. These checks do not replace Lemmas 9, 10, 11, 12, and 13. The script `conditional_chen_source_audit.py` checks source structure, references, and agreement of the current manuscript and notes with the saved certificates. It records input hashes for reproducibility; it does not establish the analytic dependencies or compile the manuscript. Neither the scripts nor successful compilation mechanically verify the analytic lemmas. The dependencies to check in a research review are:

1. The Rosser construction and main-mass estimates in BJS Proposition 13, Table 1, and Nathanson Theorem 9.3. The uniform product condition in Lemma 7 is verified at every endpoint. Lemma 8 tightens that condition by a complete finite computation and an unconditional analytic tail. Lemma 9 derives the finite medium-prime mass identity directly by first failure, and Lemma 16 proves the presieving step and its coefficient and support bounds. Lemma 11 permits independent exponents for disjoint degree groups. Lemma 12 uses a nonnegative polynomial majorant and finite weighted Cauchy–Schwarz; optimization follows complete-prime domination. Lemma 13 retains the preceding-pass constraint and uses a nonnegative tilted square with a finite weighted Gram matrix. None changes the weights. Lemma 15 sharpens their support bound without changing the coefficients, their defining level, or their finite gap estimates.
2. The AP and bilinear estimates in the accessible BS preprint. Their constants are used directly; its specialized three-term threshold formulas and normalization are not used.
3. BJS Lemma 16, JS Lemma 3.2, Lee–Nosal Corollary 1.3, the square-free reciprocal totient bound, Robin’s ω bound, and Nathanson’s combinatorial inequality.
4. Theorem 23 retains the unspecified C, X in (34); no effective EH constants have been supplied.
5. Proposition 25 retains the same missing numerical data, with a weaker signed input proved sufficient in Lemma 24. Its smaller budgets do not supply that input or bypass the product and support conditions. The cited weighted distribution results require their stated coefficient classes and residue uniformity.
6. The finite distribution hypothesis in (QEH_B) is refuted by Proposition 3; there is no full-bridge theorem.

The old floating-point probe is exploratory and is superseded by this ledger. No new unproved minorant module is hidden inside the retained cost calculation, but its distribution premise is impossible. This distinguishes positive cost arithmetic from a viable conditional research result.

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